Order Bias in Performance Evaluation: An Experiment with Sequential Contributors*

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Friday 2nd October, 2026

Abstract

Managers allocate jobs, positions, tasks and rewards based on performance evaluations. When evaluations respond to irrelevant features of production, personnel decisions may become inefficient. We test whether the sequential order of contributions to task affects evaluations using 1,811 evaluators acting as managers. Managers watch videos of workers contributing sequentially to a common output and make an incentivized hiring decision. Workers' positions in the sequence are randomly assigned and uninformative about performance. We find a systematic order bias favoring first contributors: they are hired 52.5% of the time, compared with 18.1% for last contributors. The resulting hiring decisions are also inefficient: selected workers are, on average, 38.4% less productive than the most productive available worker. The order bias remains after controlling for perceived contribution, perceived difficulty associated with each position, and actual productivity, and persists across seven additional treatments varying information and incentives.

JEL Classification: C90, D23, D91, L23, M50
Keywords: credit attribution, order effects, performance evaluation, hiring

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*We would like to thank Laila Al-Eisawi, Marielle Caballero, and Amr Yakout for excellent research assistance. Larbi Aloui, Florian Engl, Urs Fischbacher, Alex Gotthard-Real, Pedro Rey-Biel, and Robert Stuber offered very valuable suggestions. The authors are grateful for support from the Social Science Experimental Laboratory at NYU Abu Dhabi and the Center for Behavioral Institutional Design and Tamkeen under the NYUAD Research Institute award for Project CG005.

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1 Introduction

In many firms and production processes, outputs are produced through a sequence of individual contributions. Managers often have to decide who deserves credit for the final result of joint work often with incomplete information or subjective measures of individual performance (Frederiksen et al. 2020; Sarsons et al. 2021; Castilla 2025). These judgments affect hiring and who receives opportunities or rewards that impact how effectively organizations allocate talent (Minni 2026). When evaluators rely on incomplete or uninformative signals of productivity, productive workers may be overlooked and personnel decisions may become less efficient (Hoffman et al. 2018; Benson et al. 2019).

In this paper, we study whether the order of contributions systematically affects performance evaluation. Specifically, we test whether contributors who act earlier in a production sequence are assigned more credit and are more likely to be hired than contributors who act later, even when sequential position is unrelated to their performance. We refer to such an effect as *order bias in performance evaluation*.

Empirically isolating order bias is challenging. In most real-world settings, contributors differ in ability, effort, workload, task difficulty, and organizational hierarchy status. Moreover, earlier work can alter the opportunities or constraints faced by later contributors and thereby affect their productivity. Observed differences in evaluation may therefore reflect genuine differences in performance rather than order bias. A credible test requires a setting in which sequential position is assigned independently of workers' performance and in which observable and perceived dimensions of performance can be measured.

To overcome these challenges, we designed an experiment with two distinct groups of participants: *builders*, who perform real-effort production tasks in the laboratory, and evaluators, who subsequently assess their performance. Each builder first assembles a

Lego model independently, providing us with an individual measure of performance. They then work in groups of three to assemble a different Lego model sequentially while being recorded on video. Builders are randomly assigned to positions in the production sequence. The first builder begins the model, the middle builder continues its construction, and the last builder completes it. The production task is divided by the experimenter into three fixed parts. The first and last builders complete the two larger parts of the model, which are broadly comparable in size, while the middle builder completes a smaller part between them. Random assignment therefore makes builder identity and prior performance in the individual task independent of their sequential position in the production sequence.

We then recruit a separate sample of uninvolved third-party evaluators (n=1,811), hereafter *managers*, who watch the recorded production process and choose one of the three builders to hire. Their hiring decision is incentivized: they earn a higher payoff for selecting a builder who performed well (low completion time) when assembling the model in the individual building task. Managers must therefore infer performance in the individual task from the observed sequential production process. We use this incentivized hiring decision as our primary behavioral outcome. We also asked managers which builder they considered most responsible for the group outcome, allowing us to assess the connection between hiring and credit attribution.

The BASELINE treatment contains two outcome conditions. Managers observe a production group that is described as either having *succeeded* or *failed* at completing the task sufficiently quickly. Pooling the two outcome conditions, we find a sizable and robust first-builder advantage: managers hire first builders 52.5% of the time, compared with 18.1% for last builders. The advantage is present following both successful and failed outcomes, although it is larger following failure. Sequential position provides no information about performance in the separate individual task because builders are randomly assigned to positions. Moreover, in the observed sequential production task, a one minute reduction in any builder's completion time reduces the group's total com-

pletion time by exactly one minute, regardless of whether it occurs at the beginning or at the end of the sequence. The hiring gap therefore provides initial evidence of an order bias in performance evaluation. Importantly, the advantage favoring the first builder remains after controlling for managers' perceptions of individual contribution, perceived task difficulty, and actual observed productivity in the task. Hiring choices move closely with managers' responsibility judgments, providing confirming evidence that the incentivized hiring decision is connected to managers' attribution of credit and blame for the final outcome.

The hiring decisions are also materially costly and lead to substantial ex post productivity losses. Using productivity in the observed production task as a benchmark, the builder selected by managers in BASELINE produces, on average, 4.7 fewer pieces per minute than the most productive available builder. This gap corresponds to 38.4% of the highest available productivity, indicating that the personnel decisions in which the first-builder advantage arises can be associated with economically meaningful productivity consequences.

We then conducted seven additional treatments to distinguish the first-builder advantage from alternative explanations and characterize the boundaries of the order bias. First, motivated by literature on responsibility and outcome attribution that suggests the presence and timing of outcome information can affect causality judgments[1], we vary whether and when managers learn the group's outcome. The first-builder advantage persists when the outcome is disclosed before the production process is observed and when no outcome information is provided. Thus, the effect does not require managers to receive an explicitly stated outcome. Second, we reduce ambiguity about the object of evaluation by rewarding managers for selecting the builder with the highest observed productivity, measured as correct pieces assembled per minute, and vary whether this criterion is disclosed before or after the production video. The first-builder advantage persists even when managers know in advance exactly which dimension of performance

[1]We provide a discussion of the existing literature in Section 2.

will determine their payoff. Finally, we introduce features of consequential personnel decisions: the hiring choice can affect the worker's payoff, the manager's own payoff is tied to group performance, or managers can be rewarded for deliberating before making their decision. None of these changes eliminates the first-builder advantage, which remains positive and statistically significant in all seven additional treatments.

Taken together, our contribution is to document sequential position as a distinct source of distortion in performance evaluation. Existing research emphasizes imperfect signals of worker performance, subjective managerial assessments, and the difficulty of assigning individual credit for a common output. We show that evaluators systematically favor earlier contributors when builders are randomly assigned to positions and that the pattern remains after accounting for observed productivity and perceived dimensions of performance. The additional treatments show that the first-builder advantage does not depend on the timing or availability of outcome information or on uncertainty about the relevant hiring criterion, and that it persists when the decision has payoff consequences and when managers are rewarded for deliberation. We interpret this collection of evidence as documenting an order bias in performance evaluation. By connecting this pattern to an incentivized personnel decision and substantial ex post productivity shortfalls, the paper links order effects in judgment to personnel selection and organizational efficiency.

The remainder of the article is structured as follows. Section 2 discusses the related literature. Section 3 presents the experimental design. Sections 4 and 5 report the results. Section 6 concludes.

2 Related Literature

This paper contributes to three related literatures: research on personnel evaluation and worker selection, research on the attribution of credit and responsibility in joint

production, and research on order effects in judgment. We connect these literatures by identifying sequential position as a distinct source of order bias in performance evaluation. Managers systematically favor first contributors in an incentivized hiring decision even though builders are randomly assigned to positions and position is uninformative about performance relevant for the hiring incentive.

The first literature studies how organizations evaluate and allocate workers when objective measures of individual performance are incomplete. Firms often combine objective performance measures with subjective assessments because observable metrics may only imperfectly capture a worker's contribution (Baker et al. 1994). Subjective evaluation, however, can introduce favoritism, evaluator heterogeneity, and arbitrary variation in rewards and assignments (Prendergast and Topel 1996; Frederiksen et al. 2020). Related empirical work shows that incomplete signals and managerial discretion can distort hiring and promotion decisions (Pallais 2014; Hoffman et al. 2018; Benson et al. 2019), while managers can have persistent effects on workers' allocation, performance, and career progression (Minni 2026). A related theoretical and experimental literature examines incentive provision and team organization when production occurs sequentially across contributors, including how the ordering of heterogeneous team members affects productivity (Winter 2006; Collins 2016). We contribute to this work by showing that sequential position can systematically influence worker selection even when it contains no information about the individual performance relevant for the hiring incentive.

The second literature studies how evaluators assign individual credit when a common output reflects the work of multiple contributors. Sarsons et al. (2021) show that biases in the recognition of group work can affect consequential career outcomes and experimentally demonstrate that evaluators may allocate credit differently even after observable performance is held constant. Other work documents heterogeneity in norms for allocating collective credit and self-serving assessments of individual contributions when objective attribution is difficult (Shen and Barabási 2014; Herz et al.

2020). Our experiment identifies a distinct source of attributional distortion. Rather than varying a contributor's identity or relying on contributors' reports of their own inputs, we examine whether evaluations by uninvolved third parties depend on where a contribution occurs within a production sequence. We also connect the attribution problem to an incentivized hiring decision. By eliciting both hiring choices and responsibility judgments, we show that personnel selection is systematically associated with managers' attribution of credit and blame for the common outcome.

This attribution literature is closely related to work on causal and responsibility judgments. Research in psychology studies how individuals select particular events as causes within multi-stage processes and emphasizes perceived changes in the probability of the final outcome (Spellman 1997; Henne et al. 2021). In economics, studies of sequential collective decisions show that agents who are pivotal for an outcome may receive disproportionate blame or reward (Bartling et al. 2015; Bhatia et al. 2024), while theoretical work formalizes how counterfactual pivotality shapes responsibility judgments (Engl 2022). Our setting differs from environments in which one position has a mechanically greater effect on the outcome. Each builder's completion time enters the group's total completion time additively: a one minute reduction in any builder's completion time reduces the group total by exactly one minute, regardless of where that builder appears in the sequence. Moreover, the first-builder advantage persists when managers receive no stated success or failure outcome. An explicitly announced group outcome is therefore not necessary for sequential position to affect evaluation.

The third literature examines how sequential context affects judgment. A large body of research documents primacy and recency effects in information processing, with their direction depending on the task and on how evidence is presented (Hogarth and Einhorn 1992). In economics, Chen et al. (2016) shows that decisions in several high-stakes environments depend on preceding decisions even after accounting for the merits of the current case. More closely related, Radbruch and Schiprowski (2025) documents that assessments in high-stakes interview sequences depend on the quality and

ordering of other candidates evaluated by the same decision maker. Our setting differs because the relevant sequence is not a series of independent cases or successive pieces of information about the same object. Managers instead observe different individuals contributing to a common output, and the relevant position is the location of each individual's contribution within that production process. By varying the timing of outcome information and the disclosure of the hiring criterion, we examine whether the first-builder advantage depends on the outcome being announced immediately after the last contribution or on uncertainty about which dimension of performance managers should evaluate.

Taken together, these literatures establish that personnel decisions can be distorted by imperfect performance signals, that assigning individual credit for joint output is difficult, and that judgments can depend on sequential context. We connect these insights by identifying sequential position as a consequential influence on the evaluation and selection of contributors to a common output.

3 Experimental Design

The experiment consists of two parts involving distinct participant samples. In Part 1, participants whom we call *builders* complete an individual real effort task in the laboratory. Subsequently they participate in another real effort task conducted in groups in which contributions to the task are sequential. In Part 2, another set of participants recruited online serve as uninvolved third-party evaluators, whom we call *managers*.[2] Each manager watches one recorded sequential production process and makes incentivized evaluations of the three builders. In the next subsection, we first describe the procedures of our BASELINE treatment followed by seven additional treatment varia-

[2]Related experiments likewise use uninvolved third parties to evaluate production or make decisions affecting workers (see e.g., Almås et al. 2020; Cappelen et al. 2023, 2025). Here, managers make an incentivized hiring decision after directly observing the production process rather than allocating earnings based only on summary information.

tions.

3.1 Part 1: The Building Task

The building stage was conducted at the Social Science Experimental Laboratory at New York University Abu Dhabi. We recruited builders from the laboratory's registered participant pool; all were undergraduate students. Sessions were conducted with three builders at a time and lasted approximately 30 to 40 minutes. Upon arrival, builders received general instructions and completed two real effort tasks involving the assembly of Lego models from instruction manuals. They first completed an individual task independently. After all three builders had finished, they received separate instructions for the sequential group production task. In both tasks, their objective was to assemble the model correctly and as quickly as possible. The complete instructions are reproduced in Appendix A.

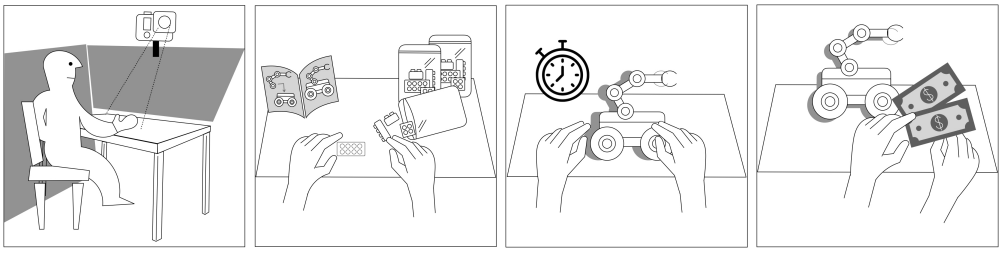
In the individual task, each builder assembled the same Lego model. The task was not recorded, and we retained each builder's completion time as a measure of individual performance. Builders received a fixed payment of 50 AED, approximately 13.6 US dollars, and an additional payment of 25 AED if their completion time placed them among the fastest 50 percent of builders in the individual task.

After completing the individual task, the three builders in a session assembled a different Lego model through a sequence of three contributions, our goal being to create a temporal distance between the first and last builders. The model was divided into three parts, with the pieces required for each part placed in a separate bag. Our intended allocation gave the first and last builders the two larger and broadly comparable parts, while the middle builder completed a smaller part between them. In implementing this allocation, the boundaries between parts also had to coincide with completed steps in the Lego instruction manual. This ensured that each builder began at the start of a new instruction step rather than midway through one. The resulting bags contained 39, 23,

and 32 pieces, respectively. The first and last builders therefore received the two larger assignments, which were closer in size to each other than either was to the middle assignment. Builders were randomly assigned to the first, middle, or last position. The first builder began the model using Bag 1, the middle builder continued its construction using Bag 2, and the last builder completed it using Bag 3. Builders were not informed of the number of pieces contained in each bag.

Separating the pieces into bags limited each builder's search to the components assigned to that part of the task. Because the parts differed in both the number and configuration of pieces, we measure each builder's observed productivity as the number of correct pieces assembled per minute and later elicit managers' perceptions of the contribution and difficulty associated with each part.

The sequential task took place in a designated cubicle and was recorded on video. The camera showed the builders' hands and the evolving Lego model but did not show their faces. The timer was paused when one builder completed a part and restarted when the next builder began, so the group's total completion time was the sum of the three active building times. Figure 1 summarizes the procedure.



a **b** **c** **d**

Figure 1 Sequential Building Task. (a) The builders sit in a cubicle, where a camera records their hands while building a Lego set following an instruction manual. (b) The Lego pieces are divided into three bags, one corresponding to each builder's part of the production sequence. (c) The timer is stopped when builders complete their parts and restarted when the next one starts. (d) A monetary bonus is paid if the group is among the fastest, as explained in the text.

A group's outcome depended on its total completion time relative to the completion times of the other groups. We used two bonus thresholds: under one threshold, a group succeeded if its completion time was among the fastest 50 percent of groups; under

the other, it succeeded if its completion time was among the fastest 25 percent. This procedure generated three production videos that could truthfully be presented as a success under the 50 percent threshold and as a failure under the 25 percent threshold, allowing us to vary the stated outcome while holding the observed production process fixed.[3]

Builders were informed that a separate sample of participants would evaluate a sped-up version of the recorded production process and that, depending on the treatment, the builder selected by an evaluator could receive an additional payment.

3.2 Part 2: Manager Evaluation and Hiring

We recruited a final sample of $1{,}811$ participants through Prolific to serve as third-party evaluators, whom we refer to as *managers*. Managers received a fixed participation payment of $\$2$ and could earn additional bonuses depending on their decisions and treatment assignment. The average participation time for managers was 8 minutes. Managers were randomly assigned to a success or failure condition, and within a condition they were randomly assigned to one of the production videos. Each manager watched one video showing three builders assembling a Lego model sequentially and then answered questions about the production process. The experimental instructions described the builders as a team whose success depended on its total completion time. In treatments that provided outcome information, managers were randomly assigned to a condition in which the production process was presented as either a *success* or a *failure*.

Managers first answered attention questions about the video. The survey then elicited

[3] We recorded nine three-builder groups and ranked them by total completion time. The two fastest groups qualified under the 25 percent threshold and were therefore presented only as successes. The next three groups qualified under the 50 percent threshold but not under the 25 percent threshold and were presented under both outcomes. The remaining four groups did not qualify under either threshold. We excluded the two slowest videos because their unusually long duration would have made the online evaluation task impractical; the other two were presented only as failures.

their evaluations of the builders. Their primary task was to choose one builder to hire. Before making this decision, managers learned that each builder had also completed a separate Lego model individually, but they did not observe performance in that task. In the BASELINE treatment, managers earned an additional $2 if the selected builder had been one of the fastest performers in the separate individual task. Because managers did not observe the builders' individual task completion times, they had to infer which builder met this criterion from the observed sequential production process. The hiring decision did not affect builders' payoffs in BASELINE.

We also asked managers which builder they considered most responsible for the group's outcome. In success conditions, the question elicited responsibility for completing the task sufficiently quickly; in failure conditions, it elicited responsibility for not doing so. This stated judgment provides a secondary outcome that allows us to examine the relationship between the incentivized hiring decision and managers' attribution of credit and blame.

Managers then estimated the number of pieces placed by each builder. They were not informed of the true distribution of pieces across positions. Managers earned an additional bonus of $1 if their estimates were within five pieces of the corresponding correct amount. These responses provide measures of perceived contribution and of managers' attention to and recall of the production process. We also asked managers which of the three parts they considered the most difficult.

Managers provided an open-ended explanation for their hiring decision. These responses allow us to examine which features of the observed process managers invoke when explaining their choices, including speed, task difficulty, amount completed, focus, and the perceived importance of a builder's position. The coding protocol for these responses is reported in Appendix D.

To examine whether managers' ranking of earlier contributors persisted beyond their initial choice, we also elicited a second incentivized hiring decision. After the initial

selection, the chosen builder was removed from the choice set and the manager selected one of the two remaining builders. The second decision used the same treatment specific performance criterion as the first, and the computer randomly determined whether it would generate an additional hiring bonus.

Finally, managers reported their gender, age, and educational attainment. We use these demographic characteristics to assess balance across the randomized treatment assignments (see Appendix Table B2). The complete manager instructions, including treatment specific wording and the ordering of the survey modules, are reproduced in Appendix A.

3.3 Treatments

The experiment contains eight treatments organized into four blocks. Table 1 summarizes the design. Block 1 contains the BASELINE treatment described above. The remaining blocks vary the timing and availability of outcome information, the performance criterion determining the hiring bonus, and the decision environment in which managers make their hiring choice.

Block 2 varies outcome information. In PREINFO, managers learn whether the group succeeded or failed before watching the production video rather than after it. This treatment changes the timing of the outcome information while preserving its content. In NOINFO, managers watch the production process without receiving information about whether the group succeeded or failed. Other aspects of the hiring decision remain as in BASELINE.

Block 3 changes the performance criterion used to determine the manager's hiring bonus. In BASELINE, managers earned an additional $2 for hiring a builder who had been one of the fastest performers in the separate individual task, which managers did not observe. They therefore had to infer performance in the individual task from the

Treatment	Outcome Information	Change relative to BASELINE	Sample
Block 1: Baseline			
BASELINE	Post-video	Benchmark: managers earn a bonus for hiring a builder who was among the fastest performers in the separate individual task	n=564
Block 2: Outcome Information			
PREINFO	Pre-video	Moves success/failure information from after the video to before the video	n=324
NOINFO	No information	Removes success/failure information from the manager's information set	n=199
Block 3: Reducing Decision Complexity			
OBJECTIVE	Post-video	Changes the hiring criterion to the highest observed productivity in the production task; criterion disclosed after viewing	n=144
PREOBJECTIVE	Pre-video	Changes the hiring criterion to the highest observed productivity in the production task; criterion disclosed before viewing	n=149
Block 4: Managerial Decision Environment			
WAGES	Post-video	Adds a probabilistic bonus for the selected worker	n=143
STAKES	Post-video	Makes the manager's payoff contingent on group success or failure	n=135
THINK	Post-video	Adds an incentive to write a justification before making the hiring decision	n=153
Total			n=1,811

Table 1 Experimental Design and Treatments.

All treatments except NOINFO include two outcome conditions, success and failure. The table reports the main experimental variation in each treatment relative to the BASELINE.

recorded sequential production process. In OBJECTIVE, the criterion instead became the builder's performance in the observed production task: managers earned the bonus for selecting the builder with the highest productivity, measured as the number of correct pieces assembled per minute. This criterion was disclosed after managers had watched the video, as in the BASELINE. In PREOBJECTIVE, managers received the same criterion before watching the video. Both treatments therefore replace an inference about performance in an unobserved task with an objectively defined criterion that can be measured from the observed task. Comparing the two treatments further allows us to examine whether knowing the criterion in advance, and therefore being able to direct attention toward it while watching the video, attenuates the first-builder advantage.

Block 4 changes the consequences and deliberative environment of the hiring decision. In WAGES, the hired builder receives a bonus of $1 with 10 percent probability. In STAKES, the manager's own payoff depends on the group outcome: managers receive an additional $1 following success and lose an initially endowed $1 following failure. In THINK, managers receive an additional $1 for explaining their reasoning before entering the hiring decision, independently of which builder they select.

3.4 Discussion of the Experimental Design

The design combines control over the assignment of builders to positions with variation in features of the evaluation environment that may generate or reduce an order effect. In BASELINE, random assignment makes builders' prior performance in the individual task independent of their position in the sequence. Although the number of pieces was not identical across positions, the first and last builders completed the two larger and more comparable parts of the task. We therefore measure productivity in the observed production task and account for managers' perceptions of contribution and task difficulty in the analysis, allowing us to examine whether these observed or perceived differences account for the first-builder advantage. Managers observe the se-

quential production process but do not observe performance in the separate individual task relevant for their hiring bonus.

The additional treatments examine whether the order bias depends on specific features of this environment or reflects a broader distortion in evaluation. The treatments in Block 2 vary whether managers receive information about success or failure and when that information becomes available. The treatments in Block 3 remove the need to infer performance in an unobserved task and, in PREOBJECTIVE, tell managers before viewing precisely which performance measure will determine their payoff. The treatments in Block 4 give managers a stake in the group outcome, allow their choices to affect builders' earnings, or reward them for explaining their reasoning before deciding.

4 Results

This section focuses on the BASELINE treatment and proceeds in four steps. We first describe the econometric strategy, a conditional logit framework, used to analyze managers' hiring decisions. We then document the first-builder advantage separately following success and failure and examine how hiring relates to managers' responsibility judgments. Third, we assess whether observed productivity in the production task and managers' perceptions of contribution and task difficulty account for the first-builder advantage, and whether the advantage is concentrated among managers who estimate contributions less accurately. Finally, we quantify the ex post productivity shortfall associated with managers' hiring decisions.

4.1 Empirical Strategy

We study hiring decisions in a setting where each manager selects exactly one builder from a set of three alternatives. The mutually exclusive nature of the outcome and the presence of builder specific attributes motivate a discrete choice approach. We therefore

estimate conditional logit models in which hiring decisions are driven by the relative utility of each builder within a given choice set.

Let i index managers and $j \in \{1, 2, 3\}$ index builders. The utility that manager i derives from selecting builder j is given by

$$U_{ij} = \alpha_1 B1_j + \alpha_2 B2_j + X_{ij}'\beta + \varepsilon_{ij},$$

where $B1_j$ and $B2_j$ are indicators for the first and middle builders in the sequence, respectively. The last builder is the omitted category. The vector X_{ij} includes builder-specific controls, and ε_{ij} is an idiosyncratic shock.

Under the standard Type I extreme value assumption, the probability that manager i hires builder j is

$$\Pr(Hire_{ij} = 1) = \frac{\exp\left(\alpha_1 B1_j + \alpha_2 B2_j + X_{ij}'\beta\right)}{\sum_{k=1}^{3} \exp\left(\alpha_1 B1_k + \alpha_2 B2_k + X_{ik}'\beta\right)}.$$

The main coefficient of interest is α_1. Since the omitted category is the last builder, a positive coefficient on $B1_j$ indicates that the first builder is more likely to be hired than the last builder, conditional on the manager's choice set and the included controls that vary across builders. The coefficient α_2 gives the corresponding difference in conditional log-odds between the middle and last builders. Exponentiated coefficients can be interpreted as odds ratios relative to the last builder.

We estimate three nested specifications. The first includes only the indicators for sequential position. The second additionally controls for perceived contribution and perceived task difficulty. Perceived contribution is the share of pieces that the manager attributes to each builder, while perceived task difficulty is measured by whether the manager identifies the part completed by the corresponding builder as the most difficult. The third specification also includes observed productivity in the production task,

measured as the number of correct pieces assembled per minute in the recorded sequential task. These covariates allow us to assess whether perceived contribution, perceived task difficulty, and productivity in the observed production task statistically account for the relationship between sequential position and hiring.

Variables that are constant across the three builders in a manager's choice set, including the treatment and stated outcome condition, are absorbed by the conditional likelihood. We therefore estimate the BASELINE models separately following success and failure, allowing the position and builder level covariate coefficients to differ across outcome conditions. Standard errors are clustered at the manager level to account for dependence across the three alternatives within each manager's choice set. We report two-sided p-values and use the conventional 10, 5, and 1 percent significance thresholds in the tables.

4.2 Baseline Hiring Results

In the BASELINE treatment, first builders are hired 52.5% of the time and last builders 18.1% of the time (Appendix Table B1). Figure 2 shows that the first-builder advantage appears under both outcome conditions. Following failure, 54.5% of managers hire the first builder and 12.8% hire the last builder. Following success, the corresponding rates are 49.8% and 25.1%. The first-builder advantage is therefore present following both outcomes and is larger following failure. Observed manager characteristics are balanced across the two outcome conditions (Appendix Table B3).

The conditional logit estimates in Table 2 confirm this pattern. In the position only specifications reported in Columns **I** and **IV**, the coefficient on the first-builder indicator (B1) is 1.45 following failure and 0.69 following success, with $p < 0.001$ in both cases. These estimates imply that the odds of hiring the first rather than the last builder are 4.27 times as high following failure and 1.98 times as high following success. The equality tests also reject the null hypothesis that the coefficients on the first and middle

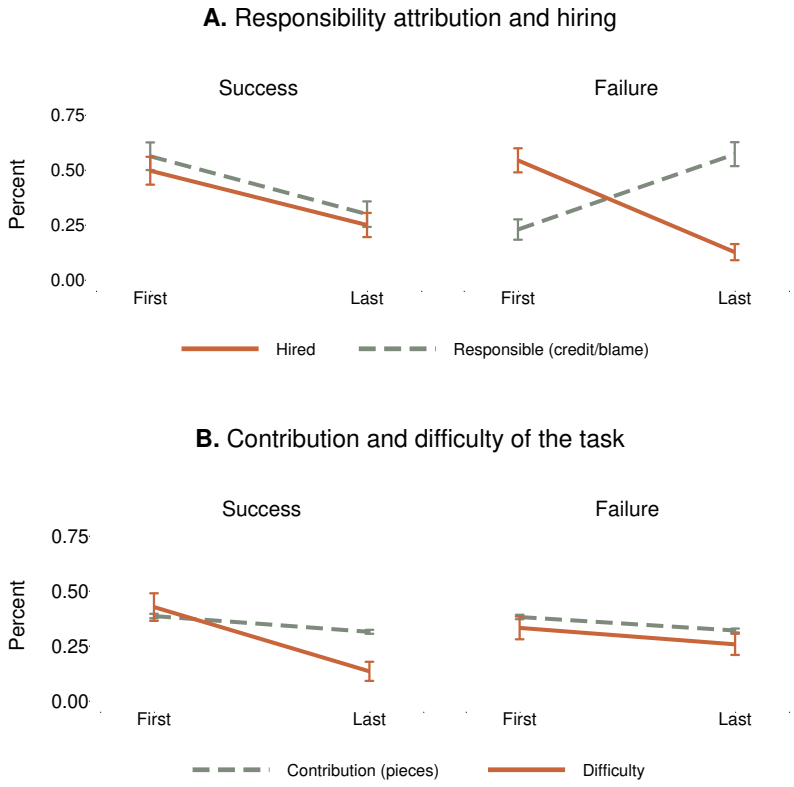


Figure 2 Main Experimental Outcomes: BASELINE. *Panel A:* The solid line displays the fraction of managers who hired the first or last builder. The dashed line displays the fraction of managers who attributed responsibility for the group outcome to the first or last builder. *Panel B:* The dashed line displays perceived contribution, measured as the manager's estimated share of pieces placed by each builder. The solid line displays perceived task difficulty, measured as the fraction of managers who identified the part of the task completed by the corresponding builder as the most difficult. Vertical bars report 95 percent confidence intervals.

builders are equal under either outcome ($p < 0.001$), indicating that the first builder is favored not only over the last builder but also over the middle builder.

For a subset of production videos, the same production process was presented to different managers as either a success or a failure. This allows us to examine how the stated outcome changes hiring while holding fixed the builders, their positions, and everything managers observe in the video. Restricting the sample to these production videos, the first-builder advantage is 37.9 percentage points following failure and 17.1 percentage points following success. The difference of 20.8 percentage points is statistically significant ($p = 0.019$). Relative to failure, presenting the same video as a success

Table 2 Hiring Decisions by Outcome: Baseline Treatment

This table reports conditional logit estimates of hiring decisions in the BASELINE treatment, separately following failure and success. The dependent variable equals one if the manager hires the corresponding builder. Columns **I** and **IV** include indicators for the first and middle builders, with the last builder as the omitted category. Columns **II** and **V** additionally control for perceived contribution and perceived task difficulty. Columns **III** and **VI** also control for productivity in the observed production task. The equality test row reports the p-value from a Wald test of the null hypothesis that the coefficients on the first and middle builders are equal. Standard errors are clustered at the manager level and reported in parentheses. Each manager contributes three builder-level observations. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	FAILURE			SUCCESS		
	I	**II**	**III**	**IV**	**V**	**VI**
First builder	1.451^{***}	1.430^{***}	1.400^{***}	0.685^{***}	0.439^{***}	0.346^{**}
	(0.174)	(0.180)	(0.187)	(0.157)	(0.169)	(0.169)
Middle builder	0.940^{***}	0.929^{***}	0.864^{***}	−0.000	0.062	−0.232
	(0.184)	(0.187)	(0.224)	(0.181)	(0.187)	(0.219)
Perceived contribution		2.274^{**}	2.299^{**}		3.171^{***}	3.077^{***}
		(0.922)	(0.931)		(0.992)	(0.988)
Perceived difficulty		-0.743^{***}	-0.743^{***}		0.145	0.131
		(0.184)	(0.184)		(0.176)	(0.174)
Observed productivity			−0.012			-0.051^{*}
			(0.026)			(0.026)
Managers	321	321	321	243	243	243
Observations	963	963	963	729	729	729
p-value: First builder = Middle builder	< 0.001	0.001	0.001	< 0.001	0.026	0.003

reduces the probability of hiring the first builder by 7.9 percentage points, although this difference is not statistically significant ($p = 0.173$), and increases the probability of hiring the last builder by 12.9 percentage points ($p = 0.005$). Thus, the group outcome affects the magnitude of the first-builder advantage, but the preference for the first builder remains positive under both outcomes (Appendix Table B4).

Responsibility judgments provide complementary evidence about the interpretation of the hiring decision. Hiring and responsibility attribution are positively correlated following success (Spearman's $\rho = 0.494$, $p < 0.001$): builders credited for the successful outcome are more likely to be hired. The relationship is negative following failure ($\rho = -0.383$, $p < 0.001$): builders blamed for the failed outcome are less likely to be hired (Appendix Table B5). These systematic associations provide evidence that the incentivized hiring decision is connected to managers' attribution of credit and blame. We

nevertheless treat hiring as the primary outcome because it is incentivized and directly represents a personnel selection decision.

Perceived Contribution, Task Difficulty, and Observed Productivity

The raw hiring gap may partly reflect systematic differences in what managers observe or infer about the builders. We therefore examine whether managers attribute different amounts of work to the first and last builders, whether they perceive the parts of the task completed by those builders as differing in difficulty, and whether observed productivity in the production task predicts hiring. Our goal is to assess whether these measures statistically account for the first-builder advantage.

Columns **II–III** and **V–VI** of Table 2 add perceived contribution, perceived task difficulty, and observed productivity in the production task to the conditional logit specifications. Perceived contribution is positively and significantly associated with hiring under both outcomes. Perceived task difficulty is negatively and significantly associated with hiring following failure but is not significantly associated with hiring following success. Observed productivity in the production task is not statistically significant following failure and is negatively associated with hiring at the 10 percent level following success. Adding these measures reduces the first-builder coefficient from 1.45 to 1.40 following failure and from 0.69 to 0.35 following success. Nevertheless, the coefficient remains positive and statistically significant in both cases. Thus, perceived contribution, perceived task difficulty, and observed productivity in the production task statistically account for part, but not all, of the first-builder advantage.

Finally, we examine whether the first-builder advantage is concentrated among managers who estimate the builders' contributions relatively inaccurately. Appendix Table B6 divides managers in BASELINE according to whether the Euclidean distance between their estimated and actual contribution vectors is at or below the median (i.e., high accuracy) or above it (i.e., low accuracy). The coefficient on the first-builder indicator

is positive both among managers with greater accuracy (1.085) and among managers with lower accuracy (1.047), with $p < 0.001$ in both groups. The persistence of the effect among more accurate managers suggests that it is not driven primarily by poor recall of, or limited attention to, how much each builder contributed.

Productivity Consequences of Hiring Decisions

To assess the economic relevance of the hiring pattern, we evaluate managers' choices using productivity in the observed production task as an ex post benchmark. In the BASELINE treatment, the selected builder produces, on average, 4.7 fewer pieces per minute than the most productive available builder. This gap corresponds to 38.4% of the highest available productivity. These ex post productivity shortfalls indicate that the personnel decisions in which the first-builder advantage arises are associated with substantial productive inefficiency (Appendix Table B7).

Result 1 (First-builder advantage and its consequences) *In the* BASELINE *treatment, managers are more likely to hire the first than the last builder following both success and failure. The first-builder advantage remains after accounting for perceived contribution, perceived task difficulty, and observed productivity and it is present among managers with both relatively higher and lower contribution estimation accuracy. Managers' hiring choices are associated with an average ex post productivity shortfall of* 38.4% *of the highest available productivity.*

5 Treatment Blocks

This section turns to the additional treatment blocks to assess the scope of the first-builder advantage and distinguish it from alternative explanations. Block 2 varies when and whether managers learn the group outcome, allowing us to examine whether the effect is driven by the timing or presence of information about success and failure.

Block 3 changes the performance criterion used for hiring and the time at which it is disclosed, allowing us to examine whether the effect reflects uncertainty about which dimension of performance matters or difficulty attending to it. Block 4 changes the decision environment by introducing consequences for the selected builder, tying the manager's payoff to the group outcome, or rewarding managers for explaining their reasoning before deciding.

Section 4 established that the first-builder advantage appears following both success and failure. Our purpose here is to test whether the overall pattern changes when specific features of the evaluation problem are altered, rather than to estimate differences between success and failure within every treatment. We therefore pool the two outcome conditions within each treatment (Appendix C reports success and failure outcomes by treatment). NoInfo, which contains no stated outcome, enters the analysis as a single sample. Together, these treatments allow us to assess whether the Baseline finding depends on specific features of outcome information, the hiring criterion, incentives, or deliberation, or instead reflects a broader distortion in performance evaluation.

Figure 3 summarizes the evidence across treatments. Panel A reports the raw hiring rates for the first and last builders. Panel B reports the coefficient on the first-builder indicator from a separate conditional logit model for each treatment, with the last builder as the omitted category. The models control for perceived contribution, perceived task difficulty, and productivity in the observed production task. Appendix Table B8 reports the complete estimates and the tests comparing the first-builder coefficient in each additional treatment with the corresponding coefficient in Baseline.

5.1 Block 2: Outcome Information

Block 2 examines whether the first-builder advantage is explained by the timing or presence of information about the group outcome. This is motivated by evidence showing that the order in which information is received can systematically affect subsequent

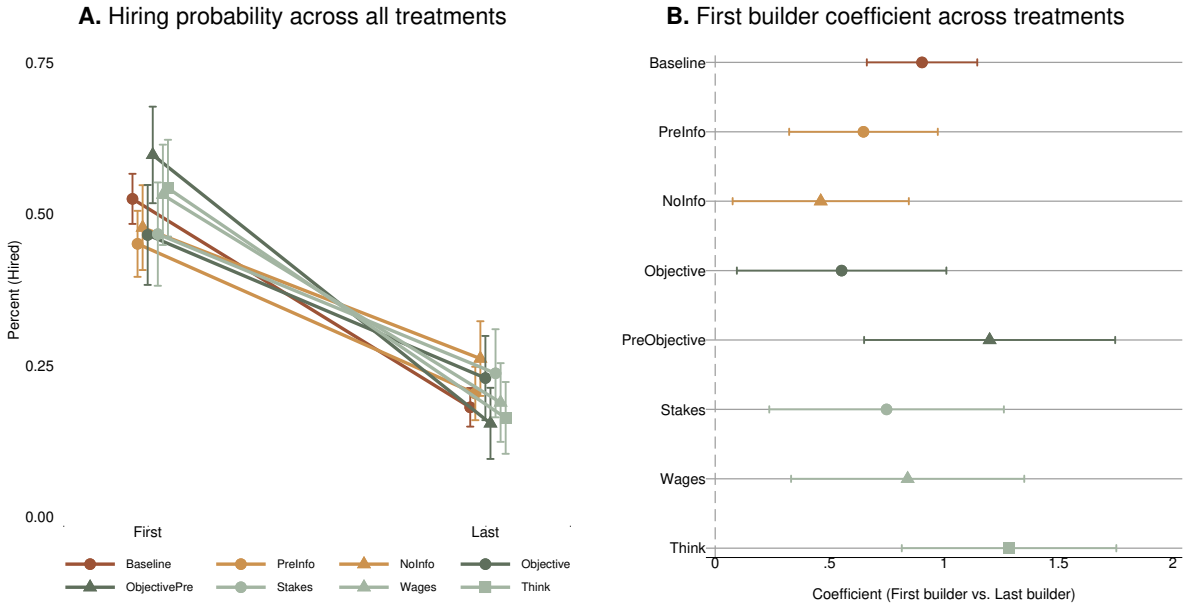


Figure 3 Hiring Patterns Across Treatments. *Panel A:* Raw hiring rates for the first and last builders in each treatment. Success and failure are pooled in all treatments that include both outcome conditions. *Panel B:* Coefficients on the first-builder indicator from separate conditional logit models for each treatment, with the last builder as the omitted category. The models control for productivity in the observed production task, perceived contribution, and perceived task difficulty. Horizontal lines report 95 percent confidence intervals.

judgments (Hogarth and Einhorn 1992). In BASELINE, managers learn whether the group succeeded or failed immediately after watching the production process. At that point, the last builder is the contributor they have most recently observed. The timing of the outcome information could therefore lead managers to connect the outcome more closely to the last builder.

Treatment PREINFO changes this timing by informing managers whether the group succeeded or failed before they watch the production process. Managers receive the same outcome information as in BASELINE, but it no longer arrives immediately after they observe the last builder. The first-builder advantage remains. First builders are hired 24.7 percentage points more often than last builders (45.1% versus 20.4%; $p < 0.001$). The estimate of the first-builder advantage does not differ statistically from the corresponding estimate in BASELINE ($p = 0.215$). Revealing the outcome before managers watch the production process therefore does not remove the preference for the first builder.

A separate possibility is that telling managers whether the group succeeded or failed leads them to search for someone to credit or blame.[4] Treatment NoInfo examines this explanation by showing managers the same production videos without stating whether the group succeeded or failed. The first-builder advantage again remains. First builders are hired 21.6 percentage points more often than last builders (47.7% versus 26.1%; $p < 0.001$). The estimate of the first-builder advantage is marginally smaller than the corresponding estimate in Baseline ($p = 0.056$).

Together, PreInfo and NoInfo show that the first-builder advantage does not require the outcome to be announced immediately after the production process or require managers to be told whether the group succeeded or failed. Managers in NoInfo may still form their own views about the quality of the production process, but an explicit success or failure outcome is not necessary for the pattern to arise.

Result 2 (Outcome information) *The first-builder advantage persists when managers learn the group outcome before observing the production process and when they receive no information about whether the group succeeded or failed. The first-builder advantage therefore does not depend on either the timing or the explicit availability of outcome information.*

5.2 Block 3: Hiring Criterion and Decision Complexity

Block 3 examines whether the first-builder advantage arises because the hiring problem in Baseline requires a difficult inference. Managers must use the observed production process to predict which builder performed well in a separate individual task that they did not observe. They may therefore rely on sequential position when performance in the individual task is difficult to infer.

[4] Work on responsibility and causal attribution emphasizes that observed outcomes can lead evaluators to consider which preceding actions were responsible for bringing them about (Spellman 1997; Henne et al. 2021; Bartling et al. 2015; Bhatia et al. 2024).

Treatment OBJECTIVE removes the need to predict performance in the individual task. Instead, the hiring bonus depends on selecting the builder with the highest productivity in the observed production task, measured as the number of correct pieces assembled per minute. Managers learn this criterion after watching the video. The criterion is therefore based on performance they directly observed.

Treatment PREOBJECTIVE uses the same hiring criterion but discloses it before managers watch the video. Managers therefore know in advance that correct pieces assembled per minute will determine their bonus and can attend to the relevant information while observing the production process.

The first-builder advantage remains in both treatments. In OBJECTIVE, first builders are hired 23.6 percentage points more often than last builders (46.5% versus 22.9%; $p < 0.001$). In PREOBJECTIVE, first builders are hired 44.3 percentage points more often than last builders (59.7% versus 15.4%; $p < 0.001$). The estimates of the first-builder advantage do not differ statistically from the corresponding estimate in BASELINE in either treatment ($p = 0.183$ and $p = 0.334$, respectively).

These results show that the first-builder advantage is not explained solely by the need to infer performance in an unobserved task. It also remains when managers know before watching the video exactly which measure of observed performance will determine their bonus. We therefore find no evidence that providing a direct hiring criterion or disclosing it in advance reduces the preference for the first builder.

Result 3 (Decision complexity) *The first-builder advantage persists when the hiring bonus depends on productivity in the observed production task and when managers know this criterion before watching the production process. The need to infer performance in an unobserved task and uncertainty about the relevant performance measure do not account for the first-builder advantage.*

5.3 Block 4: Decision Consequences and Deliberation

Block 4 examines whether the first-builder advantage persists when the hiring decision includes features common in personnel decisions. The first treatment allows the hiring choice to affect the selected builder's earnings. The second links the manager's own payoff to the group outcome. The third asks managers to explain their reasoning before making the hiring decision and pays them for it.

In WAGES, the hired builder receives a bonus of $\$1$ with 10 percent probability. The first-builder advantage remains: first builders are hired 34.2 percentage points more often than last builders (53.1% versus 18.9%; $p < 0.001$). The controlled estimate of the first-builder advantage does not differ statistically from the corresponding estimate in BASELINE ($p = 0.827$). Allowing the hiring decision to affect the selected builder's earnings therefore does not remove or significantly alter the preference for the first builder.

In STAKES, the manager's own payoff depends on the group outcome. Managers begin with an additional $\$1$, receive another $\$1$ if the group succeeds, and lose the initial $\$1$ if the group fails. First builders are hired 23.0 percentage points more often than last builders (46.7% versus 23.7%; $p < 0.001$). The estimate of the first-builder advantage does not differ statistically from the corresponding estimate in BASELINE ($p = 0.592$). The preference for the first builder therefore remains when managers have a financial stake in the group outcome.

In THINK, managers receive an additional $\$1$ for explaining their reasoning before making the hiring decision, regardless of which builder they select. This treatment gives managers an incentive to consider and state the basis for their choice before deciding. First builders are hired 37.9 percentage points more often than last builders (54.2% versus 16.3%; $p < 0.001$). The controlled estimate of the first-builder advantage is the largest among the eight treatments. Its difference from the corresponding esti-

mate in BASELINE is not statistically significant ($p = 0.157$). The data therefore provide no evidence that paying managers to explain their reasoning reduces the first-builder advantage.

Result 4 (Managerial decision environment) *The first-builder advantage persists when the hiring choice can affect the selected builder's earnings, when the manager's own payoff depends on the group outcome, and when managers are paid to explain their reasoning before making the hiring decision.*

Managers' Stated Reasons for Hiring

Next we look at managers' written stated explanations, as these provide descriptive evidence on the considerations they report using when making their hiring decisions. Pooling all treatments and outcome conditions, two categories of the seven we encoded are by far the most common stated reasons. *Speed* and *focus*, appear as motives for hiring in 63.6% and 43.2% of responses, respectively. They are also the two most frequently mentioned considerations among managers who hire the first, middle, or last builder. Each of the remaining reasons appears in no more than 15% of responses.[5]

In short, we find no systematic pattern of responses that helps distinguish when and why first contributors are preferred over last contributors. Furthermore, explicit references to order are quite rare, appearing in only 1.9% of explanations. Clearly, this does not imply that managers are unaffected by order as the results show otherwise. Instead, we interpret it as an ex post rationalization of choice justified by sensible reasons, consistent with an implicit bias in reasoning.

[5] Appendix Section D describes the coding procedure and reports the complete distribution of stated reasons.

5.4 Persistence Beyond the Initial Hiring Decision

The preceding analyses show that the first-builder advantage appears following both success and failure and persists across all treatment blocks. In this section, we take advantage of this consistency and pool all treatments together to examine whether the preference for the first builder remains visible beyond managers' initial hiring decision. The purpose is to use the second hiring decision as an overall test of managers' ranking of the builders.

After managers made their initial hiring decision, the hired builder was removed from the choice set and managers selected one of the two remaining builders.[6] We focus on the 444 managers who initially hired the middle builder. For these managers, the second decision is a direct choice between the first and last builders, and their initial choice does not already reveal how they rank these two alternatives. Of these managers, 64.6% subsequently hire the first builder rather than the last builder. This share is significantly greater than one half according to an exact binomial test ($p < 0.001$). The preference also remains after accounting for productivity in the observed production task, perceived contribution, and perceived task difficulty. In the fully controlled conditional logit model, the odds of hiring the first builder are approximately 2.73 times the odds of hiring the last builder. Appendix Table B9 reports the estimates and supporting analyses.

The second hiring decision therefore shows that, among managers whose initial choice does not reveal a preference between the first and last builders, the subsequent comparison favors the first builder. Because this analysis conditions on the initial selection of the middle builder, we interpret it as complementary evidence about managers' ranking of the remaining builders.

[6] The second hiring question was introduced during BASELINE data collection, after the first 340 managers had completed the study. It was administered to the remaining 224 BASELINE managers and to all managers in the seven additional treatments.

Result 5 (Persistence beyond the initial hiring decision) *Pooling all treatments and outcome conditions, managers who initially hire the middle builder subsequently favor the first builder over the last builder. This preference remains after accounting for perceived contribution, perceived task difficulty, and productivity in the observed production task.*

6 Conclusion

Organizations often evaluate individual contributors after observing a production process that generates a common output. This paper studies whether a contributor's position in that process systematically affects performance evaluation. Builders first completed an individual task and then assembled a different Lego model sequentially in groups of three. They were randomly assigned to the first, middle, or last position. A separate sample of $1{,}811$ managers watched one recorded production process and made an incentivized hiring decision.

We find a sizable first-builder advantage. In the BASELINE treatment, managers hire first builders 52.5% of the time and last builders 18.1% of the time. Random assignment makes builders' prior performance in the individual task independent of their position in the production sequence. The first and last builders complete the two larger and broadly comparable parts of the production task, and the first-builder advantage remains after accounting for perceived contribution, perceived task difficulty, and productivity in the observed production task. We interpret this pattern as an order bias in performance evaluation.

The first-builder advantage appears following both successful and failed outcomes, yet an explicitly stated group outcome is not necessary for the advantage to arise: it also appears in NOINFO, where managers receive no information about whether the group succeeded or failed. The second hiring decision provides complementary behavioral evidence. Pooling treatments and outcome conditions, 64.6% of managers who

initially hire the middle builder subsequently hire the first builder rather than the last builder. This preference remains after accounting for perceived contribution, perceived task difficulty, and productivity in the observed production task.

The hiring choices are also associated with economically meaningful productivity shortfalls. In BASELINE, the productivity of the hired builder is 38.4% below the highest available productivity on average. This calculation shows that the personnel decisions in which the first-builder advantage arises are associated with substantial differences in the productivity of the selected workers.

Seven additional treatments examine potential explanations for the first-builder advantage. Moving outcome information to before the production process and removing explicit success or failure information do not eliminate the effect. Changing the hiring criterion so that managers are rewarded for selecting the builder with the highest productivity in the observed production task also does not eliminate it, including when managers know this criterion before evaluation. The advantage further persists when the hiring choice can affect the hired builder's earnings, when the manager's own payoff depends on the group outcome, and when managers are paid to explain their reasoning before deciding.

Our setting is deliberately controlled and abstracts from several features of organizational production by design. The production process is short, managers observe only three contributors, and the task consists of assembling a relatively simple physical model. The first and last parts are broadly comparable but not identical, and the design limits many of the complementarities, hierarchies, and differences in responsibility that characterize organizational production. The magnitude of the first-builder advantage should therefore not be assumed to generalize directly to every production environment. These features instead allow us to establish a benchmark: even in a transparent setting with random assignment to positions, direct observation of production, and measured dimensions of performance, sequential position systematically predicts

evaluation.

Our results caution against assuming that objective performance criteria, greater consequences, or additional deliberation will by themselves remove biases based on order. Organizations that evaluate sequential contributors may need to consider explicitly how the structure and presentation of the production process influence individual assessment. Future research can examine whether the order bias persists when positions are chosen rather than assigned, when contributions involve stronger complementarities, and when managers compare several production groups rather than evaluating one group in isolation. It can also test which evaluation procedures are most effective at separating judgments of individual performance from the position at which a contribution occurs.

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Online Appendix

Order Bias in Performance Evaluation: An Experiment with Sequential Contributors

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Abstract

This Online Appendix provides supplementary materials and analyses for the paper. Appendix A reproduces the experimental instructions provided to builders and managers and provides links to the production videos. Appendix B reports descriptive statistics and supporting analyses for the main experiment. Appendix C presents the main descriptive outcomes separately for each additional treatment. Appendix D documents the coding of managers' open-ended stated reasons for hiring. Appendix E describes the relation between the preregistration and the analyses reported in the paper. Appendix F reports the original preregistered study based on production sequences constructed from independently recorded performances.

A Experimental Instructions and Video Materials

Appendix A reproduces the task and decision instructions provided to builders and managers. Section A.1 presents the written instructions for the individual and sequential production tasks, including the payment rules and the recording and subsequent evaluation of the builders' performance. Section A.2 presents the instructions for the online evaluation task, including the video protocol, attention checks, hiring and responsibility decisions, contribution and task difficulty questions, and treatment variations in outcome information, hiring criteria, and incentives. For readability, common manager instructions are reported once and treatment-specific additions or replacements are identified using italicized bracketed labels. Questions are grouped by module rather than presented as eight separate survey flows; where the order of modules differed across conditions, we state the relevant order explicitly. Section A.3 provides links to the production videos and identifies the versions used across outcome and treatment conditions.

A.1 Builder Instructions

Below we reproduce the written instructions provided to builders. Three builders participated in each session. They first completed the individual building task and then completed the sequential building task as a group. After the written instructions for Part 2 were provided, the experimenter randomly assigned one builder to each position in the sequence.

General Instructions

Welcome. You are here to participate in a study that will take between 30 and 40

minutes of your time. There are two parts in this study. We will give you instructions for part 1. Once part 1 is finished we will give you instructions for part 2.

You will receive a fixed allowance of **50 Dirhams** for your participation at the end of today's session. You can earn additional bonus allowances depending on your performance in the tasks you will be doing. The bonus allowances will be explained with the instructions for each part.

Instructions Part 1

In this first part, you will each individually build a Lego set. On your desk you can find all the pieces and the instruction manual to follow. Your task is to build the set so it looks exactly the way it is displayed on the box.

We will use a timer to record how long it takes each of you to individually build your set. Once you are done, please raise your hand so we can stop the timer for you. In addition to your participation allowance, you can earn a bonus allowance of **25 Dirhams** if your time is below the median time of all participants in this study. We will take the time it took each participant to build the lego set and put them in order from lowest to highest. If your time is among the 50% lowest times (that is, you are among the 50% fastest in building the set), you will get the bonus.

We will assign bonus allowances after all sessions are concluded.

Instructions Part 2

Now that all of you have finished building your set, we will give you the instructions for part 2. In Part 2, the three of you will build together a different Lego set as a group. The pieces of the Lego set have been divided into three bags. The first set of the pieces are in Bag 1, the next set of the pieces are in Bag 2 and the remaining pieces are in Bag 3.

The Lego set will be built in a sequence. This means that one of you will start with Bag 1. Once the first person is done, another will continue by building the pieces from Bag 2, and once this is completed the third person will finish the set by building the pieces from Bag 3.

We will record a video of the process of building the set. Only your hands will be focused on the camera. On top of that, we will record the total time it takes your group to complete the sequence and build the Lego set.

You can earn a bonus allowance of **25 Dirhams** depending on the time it takes your group to finish building the Lego set. There are two possible scenarios and your group will be randomly assigned to one of them. In scenario 1, you get the bonus allowance if your group's total time is among the 50% lowest times (that is, your group is among the 50% fastest in building the set), each of you will get the bonus. In scenario 2, you get the bonus if your group's total time is among the 25% lowest times (that is, your group is among the 25% fastest in building the set), each of you will get the bonus allowance. Each group will be randomly assigned to one of the two scenarios.

On top of that, in a separate session, we will show a speed-up version of the video to different people and ask them to evaluate each of your performances. Whoever the evaluator thinks was the best performer may receive an additional bonus allowance.

A.2 Manager Instructions

Below we reproduce the instructions shown to managers. Ordinary text reproduces wording displayed to participants. Italicized bracketed labels are editorial annotations and were not shown to participants. These labels identify the treatment or outcome condition to which a passage applied. Text described as a replacement was displayed instead of the preceding common passage. The labels *[*SUCCESS*]* and *[*FAILURE*]* identify wording shown only in the corresponding outcome condition. The label *[*NOINFO*]*

identifies wording used when managers received no information about whether the group earned the bonus. The notation *[*randomized order]* indicates that the response options were displayed in random order. We use *[page break]* to indicate a transition between survey screens and *[video displayed]* to indicate where the production video appeared.

For readability, the instructions are grouped below by survey module rather than reproduced as eight separate survey flows. The second hiring question was introduced partway through BASELINE data collection. It was shown to all subsequent BASELINE managers and to all managers in the seven additional treatments. The success and failure announcements were embedded in the production videos rather than displayed on separate survey screens. Their timing is identified below, and links to the corresponding video versions are provided in Section A.3.

Welcome and consent

WELCOME

You are participating in a study on economic decision-making. Typically, the study takes about **8 minutes** to complete.

*[*BASELINE*,* PREINFO*,* NOINFO*,* OBJECTIVE*,* PREOBJECTIVE*, and* WAGES*]*

For completing the study, you will receive **$2.00**. In addition, you will be able to earn **bonus payments** of up to **$3.00 more**. You will be paid only if you complete the entire study.

*[*THINK*: replacement for the preceding payment paragraph]*

For completing the study, you will receive **$2.00**. In addition, you will be able to earn **bonus payments** of up to **$4.00 more**. You will be paid only if you complete the entire study.

*[*STAKES*: replacement for the preceding payment paragraph]*

For completing the study, you will receive **$2.00**. In addition, you will be able to earn **bonus payments** of up to **$5.00 more**. You will be paid only if you complete the entire study.

The study is anonymous. Hence, your identity will not be revealed to others and the identity of others will not be revealed to you.

Next, you will see the instructions. **Please read the instructions carefully as they describe how your earnings are determined.**

The main task is to carefully watch a short video. You will be asked questions to confirm that you have watched it with attention. **If you answer these questions incorrectly, you will be excluded from the study and you will not be eligible for payment.**

By continuing to the next screen, you consent to participate in this study. For more details about your consent, click on "See consent form."

- See consent form

[page break]

Video instructions and pre-video treatment variations

*[*BASELINE *and all treatments except* NOINFO*]*

We asked participants in a different study to build a Lego set in a team and gave a bonus to those teams for which the total time of building the Lego set was among the fastest in the study.

The following video shows you a team of participants building the Lego set. Please watch the video carefully, as we will ask you a few attention questions about the task the participants are performing.

*[*NOINFO*: replacement for the preceding introduction]*

We asked participants in a different study to build a Lego set in a team.

The following video shows you a team of participants building the Lego set. Please watch the video carefully, as we will ask you a few attention questions about the task the participants are performing.

*[*PREOBJECTIVE*: additional information shown before the video]*

Each participant built a part of the Lego set. In each of the parts, the participant was given a different number of Lego pieces, which took each participant a certain amount of time to assemble them.

We measure the performance of a participant by their productivity, which is the total number of pieces they assembled divided by the time it took them to do so.

Later in this study, you will have the opportunity to hire one of the participants in the video. If the person you hire is the best performer, meaning the one with the highest productivity among the participants, you will earn an additional bonus of **2 USD**.

*[*STAKES*: additional information shown before the video]*

Additional bonus

We have selected you, among the participants in this study, to get an additional bonus that will start at **$1 USD**. The total of this bonus will depend on whether your team is successful or not.

If your team in the video builds the Lego in time and earns their bonus, we will give you another **$1 USD**. This means you will have **$2 USD** added to your final payment.

But, if your team fails to complete the task on time and does not earn their bonus, we will take away the **$1 USD** we gave you. So, you will not have anything added to your final payment.

*[*PREINFO*: the treatment-specific video disclosed whether the team earned its bonus before the production sequence began.]*

*[*BASELINE*,* OBJECTIVE*,* PREOBJECTIVE*,* WAGES*,* STAKES*, and* THINK*: the team outcome was disclosed at the end of the production video.]*

*[*NOINFO*: the video did not disclose whether the team earned a bonus.]*

[page break]

Please watch carefully the following video.

Note: Please pay close attention to the task participants are performing. Skipping this stage or leaving before the video ends will make you ineligible for payment.

[video displayed]

Attention questions

[page break]

*[*BASELINE*,* PREINFO*,* OBJECTIVE*,* PREOBJECTIVE*,* WAGES*,* STAKES*, and* THINK*]*

Did the participants in the team earn the bonus by successfully building the Lego set in time?

- Yes
- No

[page break]

[ALL TREATMENTS]

How many participants composed the team, meaning built the Lego set, in the video you watched? *(Hint: the number of participants is equal to the number of parts in the video.)*

- Options from 0 to 10

*[*STAKES*: outcome-contingent payment screen shown after the attention questions]*

[SUCCESS]

Your Additional bonus

Your team completed building the Lego set in time. So we have added another **$1 USD** to your initial bonus. A total bonus of **$2 USD** will be added to your final payment.

[FAILURE]

Your Additional bonus

Your team did not complete building the Lego set in time. So we have taken away the **$1 USD** initially given to you. Nothing from this bonus will be added to your final payment.

Contribution and task-difficulty questions

[page break]

[ALL TREATMENTS]

Please indicate how many pieces you think **each participant** put together. If your answer is within 5 units of the correct amount of pieces for each participant, you will receive a bonus of **1 USD**.

- Participant 1: [number of pieces]
- Participant 2: [number of pieces]
- Participant 3: [number of pieces]

- Total: [automatically calculated]

[page break]

Please indicate **how difficult** you think the different parts of building the Lego set were. *[*randomized order]*

- All three parts were equally difficult
- The first part was the most difficult
- The second part was the most difficult
- The third part was the most difficult

Responsibility judgments

*[*SUCCESS*: all treatments except* NOINFO*]*

[page break]

Which of the three participants in the team was **the most responsible for building the Lego set in time to earn the bonus**? *[*randomized order]*

- Participant 1
- Participant 2
- Participant 3

[page break]

Please tell us why you said *[Chosen participant]* was the most responsible for building the Lego set in time to earn the bonus.

*[*BASELINE*: second-choice subsample; and all non-Baseline treatments,* SUCCESS*]*

[page break]

Aside from the participant you chose before, which of the remaining two participants in the team was the other most responsible for building the Lego set in time to earn the bonus? *[*randomized order; the participant selected previously was unavailable]*

- Participant 1
- Participant 2
- Participant 3

*[*FAILURE*: all treatments except* NOINFO*]*

[page break]

Which of the three participants in the team was **the most responsible for not building the Lego set in time to earn the bonus**? *[*randomized order]*

- Participant 1
- Participant 2
- Participant 3

[page break]

Please tell us why you said *[Chosen participant]* was the most responsible for not building the Lego set in time to earn the bonus.

*[*BASELINE*: second-choice subsample; and all non-Baseline treatments,* FAILURE*]*

[page break]

Aside from the participant you chose before, which of the remaining two participants in the team was the other most responsible for not building the Lego set in time to earn the bonus? *[*randomized order; the participant selected previously was unavailable]*

- Participant 1
- Participant 2
- Participant 3

*[*NOINFO*: outcome-neutral responsibility questions]*

[page break]

Which of the three participants in the team was **the most responsible for building the Lego set**? *[*randomized order]*

- Participant 1
- Participant 2
- Participant 3

[page break]

Please tell us why you said *[Chosen participant]* was the most responsible for building the Lego set.

[page break]

Aside from the participant you chose before, which of the remaining two participants in the team was the other most responsible for building the Lego set? *[*randomized order; the participant selected previously was unavailable]*

- Participant 1
- Participant 2
- Participant 3

Hiring decision

[page break]

Hiring Decision

*[*BASELINE*,* PREINFO*,* NOINFO*,* WAGES*, and* STAKES*]*

In the video you watched, each participant built one of the three parts of the Lego set. We also asked each participant to also build a different Lego set on their own, meaning that each of them built an entire Lego set by him or herself.

You have the opportunity to **hire one of the three participants** in the video. If the person you hire has built the individual Lego as one of the fastest, you will earn an additional bonus of **2 USD**.

*[*WAGES*: additional sentence shown before the hiring choice]*

The participant you hire will receive a bonus of **1 USD** with **10% probability**.

*[*OBJECTIVE *and* PREOBJECTIVE*: replacement for the Baseline hiring criterion]*

In the video you watched, each participant built one of the three parts of the Lego set. In each of those three parts, the participant was given a different number of Lego pieces, which took each participant a certain amount of time to assemble them.

We measure the performance of a participant by their productivity, which is the total number of pieces they assembled divided by the time it took them to do so.

You have the opportunity to **hire one of the three participants** in the video. If the person you hire is the best performer, meaning the one with the highest productivity

among the three, you will earn an additional bonus of **2 USD**.

*[*THINK*: replacement for the Baseline hiring screen]*

In the video you watched, each participant built one of the three parts of the Lego set. We also asked each participant to also build a different Lego set on their own, meaning that each of them built an entire Lego set by him or herself.

You have the opportunity to **hire one of the three participants** in the video. If the person you hire has built the individual Lego as one of the fastest, you will earn an additional bonus of **2 USD**. In the next question, you will be asked to indicate which of the three participants you choose to hire.

Please describe your reasoning process behind your hiring choice. For your effort in explaining your thoughts, we will pay you an additional **1 USD**, regardless of who you hire.

[page break]

[ALL TREATMENTS]

Please indicate which of the three participants you choose to hire. *[*randomized order]*

- Participant 1
- Participant 2
- Participant 3

[All treatments except THINK*]*

[page break]

Please tell us why you hired *[Hired participant]*.

Second hiring decision

*[*BASELINE*: second-choice subsample;* PREINFO*;* NOINFO*;* WAGES*;* STAKES*; and* THINK*]*

[page break]

You have the opportunity to hire another participant and the computer will randomly decide whether you can earn another bonus of **2 USD**, if the person you hire has built the individual Lego as one of the fastest. Who would you choose? *[*randomized order; the participant selected in the first hiring decision was unavailable]*

- Participant 1
- Participant 2
- Participant 3

*[*OBJECTIVE *and* PREOBJECTIVE*: replacement for the preceding second-hiring question]*

[page break]

You have the opportunity to hire another participant and the computer will randomly decide whether you can earn another bonus of **2 USD**, if the person you hire has built the individual Lego as one of the most productive. Who would you choose? *[*randomized order; the participant selected in the first hiring decision was unavailable]*

- Participant 1
- Participant 2
- Participant 3

Final questions

*[*ALL TREATMENTS AND ALL BASELINE VERSIONS*]*

[page break]

Before concluding, we would like you to answer a final set of questions.

What is your gender?

- Male
- Female
- Other

[page break]

How old are you, in years?

[page break]

What is your highest educational degree?

- Less than high school
- High school graduate
- 2 year college/university degree
- 4 year college/university degree
- Masters degree or equivalent
- Doctorate or equivalent

Additional questions in the initial BASELINE *version*

*[*BASELINE*: initial version only]*

The initial BASELINE version also asked managers to report the perceived gender of each builder, their race or ethnicity, employment status, student status, state of residence, whether they voted in the previous election, their political identification, and their position on a seven-point liberal-conservative scale. These questions were not included in the later BASELINE version or in the seven additional treatments, were not used in the analysis and are omitted here.

End of the study

[page break]

You have reached the end of the study.

Please do not discuss the procedures or content of this study with other participants.

Your bonus payment

In the next 48 hours we will review your responses to the questions that had a bonus payment and determine whether you earn the **bonus payment**. To receive your payment, you must submit your Prolific ID on the next page.

A.3 Links to videos

The links below identify the video versions shown to managers. We report the versions used for each applicable outcome.

- **Failure:**

- https://www.youtube.com/watch?v=XyzEDlx5PUA&t=1s
- https://www.youtube.com/watch?v=7Ot2o687zNs
- https://www.youtube.com/watch?v=ah5C1FlSOM8
- https://www.youtube.com/watch?v=nOqfk1vKoHQ
- https://www.youtube.com/watch?v=qy2BYFiWgZc
- https://www.youtube.com/watch?v=wvSriQu32GA

- **Success:**
  - https://www.youtube.com/watch?v=r1LzY-4UQFk
  - https://www.youtube.com/watch?v=UkWnukazX24
  - https://www.youtube.com/watch?v=2MoPYoea7_w
  - https://www.youtube.com/watch?v=9ItjXh4SPXI
  - https://www.youtube.com/watch?v=wxcUOO_bNGM
  - https://www.youtube.com/watch?v=7v3opnWjig4
  - https://www.youtube.com/watch?v=j971ozWuj2c&t=1s

B Supporting Analyses

This appendix presents the descriptive statistics and supporting analyses in the order in which they are introduced in the main text. Table B1 reports hiring probabilities, perceived contribution, and perceived task difficulty by treatment. Table B2 assesses balance in observed manager characteristics across treatments, and Table B3 assesses balance between the failure and success conditions in BASELINE. Table B4 compares hiring decisions for production videos that were presented to different managers under both outcomes, holding the observed production process fixed. Table B5 reports the association between hiring and responsibility attribution. Table B6 examines whether the first-builder advantage is present among managers with relatively higher and lower accuracy in estimating contributions, and Table B7 reports the ex post productivity shortfall associated with managers' hiring decisions. Table B8 presents the fully controlled conditional logit estimates across treatments. Finally, Table B9 reports the pooled analysis of the second hiring decision among managers who initially hired the middle builder.

B.1 Descriptive Statistics and Baseline Analyses

Table B1 Hiring, Perceived Contribution, and Perceived Task Difficulty by Treatment

This table reports descriptive statistics for managers' hiring decisions and perceptions. For treatments with outcome information, success and failure are pooled; NoInfo enters as a single sample. Panel A reports the probability that each builder is hired. Panel B reports the mean share of pieces that managers attribute to each builder. Panel C reports the probability that managers identify the part completed by each builder as the most difficult. The probabilities in Panel C do not necessarily sum to one because managers could instead report that all three parts were equally difficult. Standard errors clustered at the manager level are reported in parentheses. Each manager contributes one observation for each builder.

	I	II	III	IV	V	VI	VII	VIII
	Baseline	PreInfo	NoInfo	Objective	PreObjective	Wages	Stakes	Think
A. Hiring probability								
First builder	0.525	0.451	0.477	0.465	0.597	0.531	0.467	0.542
	(0.021)	(0.028)	(0.036)	(0.042)	(0.040)	(0.042)	(0.043)	(0.041)
Middle builder	0.294	0.346	0.261	0.306	0.248	0.280	0.296	0.294
	(0.019)	(0.026)	(0.031)	(0.039)	(0.036)	(0.038)	(0.040)	(0.037)
Last builder	0.181	0.204	0.261	0.229	0.154	0.189	0.237	0.163
	(0.016)	(0.022)	(0.031)	(0.035)	(0.030)	(0.033)	(0.037)	(0.030)
B. Perceived contribution (share of pieces)								
First builder	0.385	0.376	0.374	0.381	0.398	0.385	0.378	0.372
	(0.004)	(0.005)	(0.006)	(0.008)	(0.006)	(0.007)	(0.007)	(0.007)
Middle builder	0.296	0.302	0.301	0.297	0.279	0.295	0.302	0.306
	(0.003)	(0.003)	(0.004)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Last builder	0.319	0.322	0.325	0.322	0.323	0.320	0.320	0.321
	(0.003)	(0.004)	(0.006)	(0.007)	(0.005)	(0.006)	(0.007)	(0.006)
C. Perceived as the most difficult part								
First builder	0.374	0.392	0.322	0.396	0.470	0.364	0.378	0.379
	(0.020)	(0.027)	(0.033)	(0.041)	(0.041)	(0.040)	(0.042)	(0.039)
Middle builder	0.135	0.123	0.181	0.174	0.134	0.175	0.163	0.124
	(0.014)	(0.018)	(0.027)	(0.032)	(0.028)	(0.032)	(0.032)	(0.027)
Last builder	0.206	0.201	0.221	0.188	0.148	0.238	0.119	0.209
	(0.017)	(0.022)	(0.030)	(0.033)	(0.029)	(0.036)	(0.028)	(0.033)
Managers	564	324	199	144	149	143	135	153

Table B2 Balance of Manager Characteristics Across Treatments

This table compares observed manager characteristics across treatments. Column **I** reports an indicator equal to one if the manager identifies as female. Column **II** reports age in years. Column **III** reports an indicator equal to one if the manager has completed at least a two-year college degree. The first row reports the mean in BASELINE. The remaining rows report the difference between each additional treatment and BASELINE. Robust standard errors are reported in parentheses. The final row reports the p-value from a joint test that the coefficients on the seven additional treatment indicators are equal to zero. Each manager contributes one observation. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	I	**II**	**III**
	FEMALE	AGE	EDUCATION
BASELINE mean	0.486^{***}	43.869^{***}	0.746^{***}
	(0.021)	(0.594)	(0.018)
PREINFO − BASELINE	0.017	0.036	−0.018
	(0.035)	(0.970)	(0.031)
NOINFO − BASELINE	0.047	0.302	0.007
	(0.041)	(1.071)	(0.036)
OBJECTIVE − BASELINE	−0.014	−1.799	−0.010
	(0.047)	(1.233)	(0.041)
PREOBJECTIVE − BASELINE	−0.070	4.641^{***}	0.005
	(0.046)	(1.320)	(0.040)
WAGES − BASELINE	0.046	-4.295^{***}	0.016
	(0.047)	(1.215)	(0.040)
STAKES − BASELINE	0.055	-3.758^{***}	−0.028
	(0.048)	(1.342)	(0.043)
THINK − BASELINE	0.044	2.824^{**}	−0.067
	(0.046)	(1.233)	(0.042)
Managers	1,811	1,811	1,811
p-value: Joint equality	0.330	< 0.001	0.795

Table B3 Balance Between Failure and Success: Baseline Treatment

This table compares manager characteristics between the failure and success conditions in BASELINE. Column **I** reports an indicator equal to one if the manager identifies as female. Column **II** reports age in years. Column **III** reports an indicator equal to one if the manager has completed at least a two-year college degree. The first row reports the mean in the failure condition. The second row reports the difference between the success and failure means, with robust standard errors in parentheses. The Success mean row reports the corresponding mean in the success condition. The final row reports the p-value from a two-sided test of equality between the failure and success means. Each manager contributes one observation.

	I	**II**	**III**
	FEMALE	AGE	EDUCATION
Failure mean	0.470	43.695	0.751
	(0.028)	(0.822)	(0.024)
Success − Failure	0.036	0.404	−0.010
	(0.043)	(1.181)	(0.037)
Success mean	0.506	44.099	0.741
Managers	564	564	564
p-value: Failure = Success	0.401	0.732	0.787

Table B4 Hiring Decisions for the Same Production Videos Under Success and Failure

This table restricts the BASELINE sample to production videos that were presented to different managers as both successful and failed. Column **I** reports the probability of hiring the first builder. Column **II** reports the probability of hiring the last builder. Column **III** reports the difference between these two probabilities. The Success − Failure row reports the change in each outcome when the same observed production process is presented as successful rather than failed. All models include production video fixed effects. The adjusted means are evaluated separately under failure and success. The equality-test row reports the p-value from a two-sided test of equality between the two outcome conditions. Robust standard errors are reported in parentheses. Each manager contributes one observation.

	I	**II**	**III**
	FIRST BUILDER HIRED	LAST BUILDER HIRED	FIRST − LAST GAP
Success − Failure	−0.079	0.129^{***}	-0.208^{**}
	(0.058)	(0.046)	(0.088)
Adjusted mean: Failure	0.506	0.126	0.379
Adjusted mean: Success	0.426	0.256	0.171
Managers	299	299	299
p-value: Failure = Success	0.173	0.005	0.019

Table B5 Association Between Hiring and Responsibility Attribution

This table reports Spearman rank correlations between the hiring indicator and the responsibility indicator for each builder, separately by treatment and outcome condition. Following failure, responsibility identifies the builder held most responsible for the unsuccessful outcome, so negative correlations indicate that blamed builders are less likely to be hired. Following success, responsibility identifies the builder credited for the successful outcome, so positive correlations indicate that credited builders are more likely to be hired. Two-sided p-values are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	I	**II**
	FAILURE	SUCCESS
BASELINE	-0.383^{***}	0.494^{***}
	(< 0.001)	(< 0.001)
PREINFO	-0.418^{***}	0.466^{***}
	(< 0.001)	(< 0.001)
OBJECTIVE	-0.338^{***}	0.449^{***}
	(< 0.001)	(< 0.001)
PREOBJECTIVE	-0.280^{***}	0.493^{***}
	(< 0.001)	(< 0.001)
WAGES	-0.375^{***}	0.556^{***}
	(< 0.001)	(< 0.001)
STAKES	-0.434^{***}	0.328^{***}
	(< 0.001)	(< 0.001)
THINK	-0.343^{***}	0.581^{***}
	(< 0.001)	(< 0.001)

Table B6 Hiring by Accuracy of Contribution Estimates: Baseline Treatment

This table reports conditional logit estimates of hiring decisions in BASELINE, pooling success and failure and separating managers according to the accuracy of their estimates of the number of pieces placed by the three builders. Accuracy is measured by the Euclidean distance between the manager's estimated contribution vector and the actual contribution vector. A smaller distance indicates greater accuracy. Column **I** includes managers whose distance is at or below the median, while Column **II** includes managers whose distance is above the median. The models include indicators for the first and middle builders, with the last builder as the omitted category. The equality-test row reports the p-value from a Wald test of the null hypothesis that the coefficients on the first and middle builders are equal. Standard errors are clustered at the manager level and reported in parentheses. Each manager contributes three builder-level observations. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	I	**II**
	AT OR BELOW MEDIAN	ABOVE MEDIAN
First builder	1.085^{***}	1.047^{***}
	(0.166)	(0.160)
Middle builder	0.586^{***}	0.386^{**}
	(0.179)	(0.178)
Median Euclidean distance	22.924	22.924
Managers	282	282
Observations	846	846
p-value: First builder = Middle builder	< 0.001	< 0.001

Table B7 Productivity Consequences of Hiring Decisions: Baseline Treatment

This table reports the ex post productivity consequences of managers' hiring decisions in BASELINE, separately following failure and success. Column **I** reports the difference, in pieces assembled per minute, between the highest productivity among the three builders and the productivity of the builder hired by the manager. Column **II** expresses this shortfall as a percentage of the highest available productivity. The Failure and Success rows report condition-specific means, with robust standard errors in parentheses. The equality-test row reports the p-value from a two-sided test of equality across outcome conditions.

	I ABSOLUTE SHORTFALL PIECES PER MINUTE	II RELATIVE SHORTFALL PERCENT OF BEST
Failure	4.564	39.219
	(0.231)	(1.484)
Success	4.814	37.374
	(0.262)	(1.716)
Overall mean	4.672	38.424
Managers	564	564
p-value: Failure = Success	0.475	0.417

B.2 Cross-Treatment and Pooled Analyses

Table B8 Hiring Decisions Across Treatments

This table reports the fully controlled conditional logit estimates of hiring decisions separately by treatment. For treatments with outcome information, success and failure are pooled; NOINFO enters as a single sample. Columns **I–VIII** follow the treatment blocks presented in the main text. The dependent variable equals one if the manager hires the corresponding builder. The models include indicators for the first and middle builders, with the last builder as the omitted category, and control for productivity in the observed production task, perceived contribution, and perceived task difficulty. The first equality-test row reports the p-value from a Wald test that the coefficients on the first and middle builders are equal. The second reports the p-value from a two-sided test comparing the first-builder coefficient in each additional treatment with the corresponding coefficient in BASELINE; the BASELINE cell is blank because it is the reference treatment. Standard errors are clustered at the manager level and reported in parentheses. Each manager contributes three builder-level observations. Sample sizes correspond to the observations included in the fully controlled specification and may be smaller than the full treatment samples when a control is missing. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	I BASELINE	II PREINFO	III NOINFO	IV OBJECTIVE	V PREOBJECTIVE	VI WAGES	VII STAKES	VIII THINK
First builder	0.904^{***}	0.648^{***}	0.461^{**}	0.552^{**}	1.199^{***}	0.841^{***}	0.749^{***}	1.284^{***}
	(0.123)	(0.166)	(0.196)	(0.233)	(0.280)	(0.260)	(0.261)	(0.239)
Middle builder	0.354^{**}	0.506^{**}	0.095	0.326	0.383	0.371	0.418	1.119^{***}
	(0.154)	(0.214)	(0.275)	(0.309)	(0.345)	(0.352)	(0.345)	(0.316)
Observed productivity	-0.032^{*}	−0.013	−0.001	−0.011	−0.047	−0.021	0.019	0.076^{*}
	(0.019)	(0.027)	(0.036)	(0.039)	(0.049)	(0.043)	(0.041)	(0.040)
Perceived contribution	2.614^{***}	1.928^{**}	3.325^{***}	3.296^{***}	2.861^{*}	3.963^{***}	2.233^{*}	3.783^{**}
	(0.658)	(0.877)	(1.137)	(1.079)	(1.546)	(1.393)	(1.292)	(1.572)
Perceived task difficulty	-0.330^{***}	0.103	−0.001	−0.155	−0.305	-0.479^{**}	-0.507^{*}	−0.229
	(0.122)	(0.153)	(0.190)	(0.210)	(0.256)	(0.239)	(0.274)	(0.251)
Managers	564	324	199	144	148	143	135	153
Observations	1,692	972	597	432	444	429	405	459
p-value: First builder = Middle builder	< 0.001	0.390	0.086	0.364	0.004	0.071	0.209	0.530
p-value: First-builder coefficient vs. BASELINE		0.215	0.056	0.183	0.334	0.827	0.592	0.157

Table B9 Second Hiring Choice After Initially Hiring the Middle Builder

This table reports conditional logit estimates of managers' second hiring decisions, pooling all treatments and outcome conditions. The sample is restricted to managers who initially hired the middle builder and completed the second hiring decision. Because the initially hired builder was removed from the choice set, these managers subsequently chose between the first and last builders. The dependent variable equals one for the builder hired in the second decision, and the last builder is the omitted category. Column **I** includes only the first-builder indicator. Column **II** additionally controls for productivity in the observed production task and perceived contribution. Column **III** also controls for perceived task difficulty. The percentage reported under Column **III** is the raw share of managers who hired the first rather than the last builder, and the associated p-value is from an exact binomial test of the null hypothesis that the two builders are selected with equal probability. The second hiring question was introduced partway through BASELINE data collection and was administered to all subsequent BASELINE managers and to all managers in the seven additional treatments. Standard errors are clustered at the manager level and reported in parentheses. Each manager contributes two builder-level observations. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	I	**II**	**III**
	MODEL 1	MODEL 2	MODEL 3
First builder	0.603^{***}	0.968^{***}	1.004^{***}
	(0.099)	(0.146)	(0.146)
Observed productivity		0.106^{***}	0.096^{***}
		(0.026)	(0.025)
Perceived contribution		-0.021	1.212
		(0.771)	(0.822)
Perceived task difficulty			-0.567^{***}
			(0.148)
Managers	444	444	444
Observations	888	888	888
Percentage hiring the first builder			64.6%
p-value: First builder share = 0.50			< 0.001

C Additional Treatment Figures

This appendix presents the main descriptive outcomes separately for each of the seven additional treatments, using the same format as the BASELINE figure reported in Figure 2. Figures C1 and C2 correspond to the treatments that vary the timing and availability of outcome information. Figures C3 and C4 correspond to the treatments that change the performance criterion used for the hiring bonus and the time at which that criterion is disclosed. Figures C5, C6, and C7 correspond to the treatments that change the decision environment. In each figure, Panel A reports hiring and responsibility judgments for the first and last builders, while Panel B reports perceived contribution and perceived task difficulty. For treatments in which managers receive information about the group outcome, the results are presented separately following success and failure. In NOINFO, managers receive no information about whether the group succeeded or failed, so the full treatment sample is presented in a single panel. The middle builder is omitted from these figures to focus on the comparison between the beginning and end of the production sequence. Vertical bars report 95 percent confidence intervals.

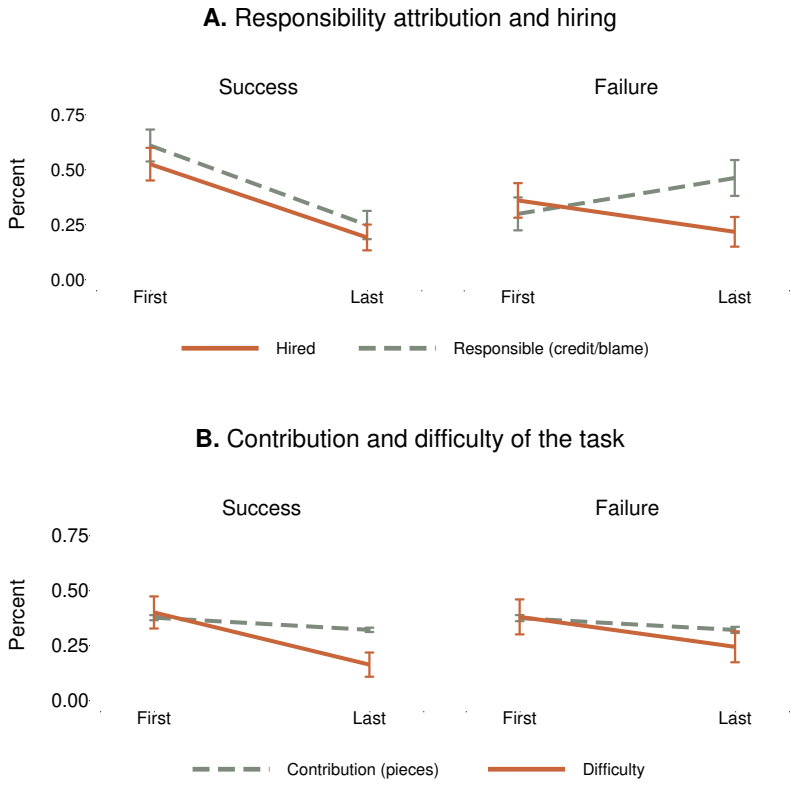


Figure C1 Main Experimental Outcomes: PREINFO. *Notes:* Panel A reports the fraction of managers who hired the first or last builder and the fraction who attributed responsibility for the group outcome to the first or last builder. Panel B reports perceived contribution, measured as the estimated share of pieces placed by each builder, and perceived task difficulty, measured as the fraction of managers who identified the part completed by the corresponding builder as the most difficult. Results are presented separately following success and failure. Vertical bars report 95 percent confidence intervals.

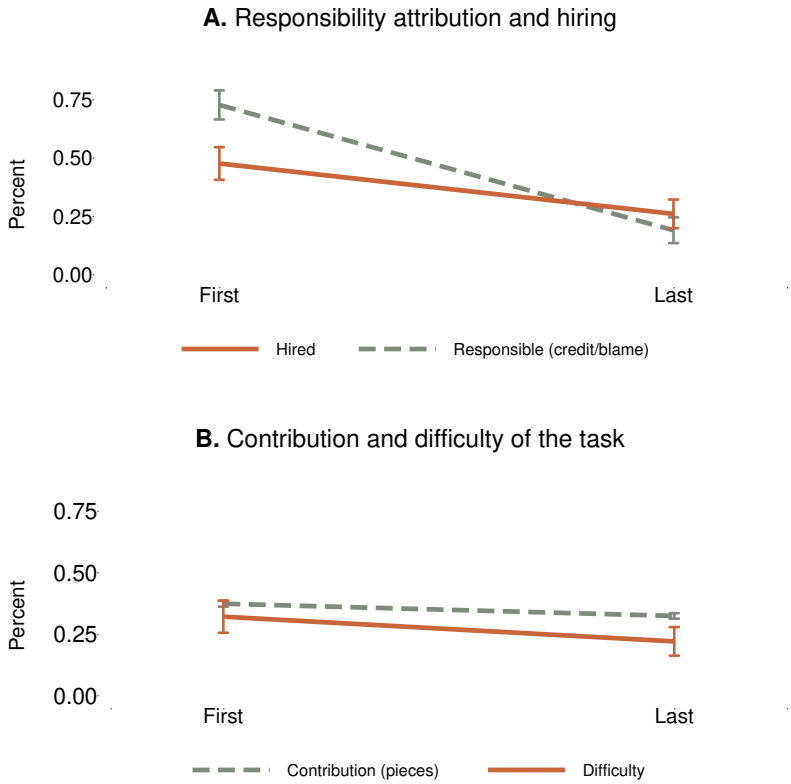


Figure C2 Main Experimental Outcomes: NoInfo. *Notes:* Panel A reports the fraction of managers who hired the first or last builder and the fraction who identified the first or last builder as most responsible for building the Lego set. Panel B reports perceived contribution, measured as the estimated share of pieces placed by each builder, and perceived task difficulty, measured as the fraction of managers who identified the part completed by the corresponding builder as the most difficult. Managers received no information about whether the group succeeded or failed, so the figure reports the full treatment sample in a single panel. Vertical bars report 95 percent confidence intervals.

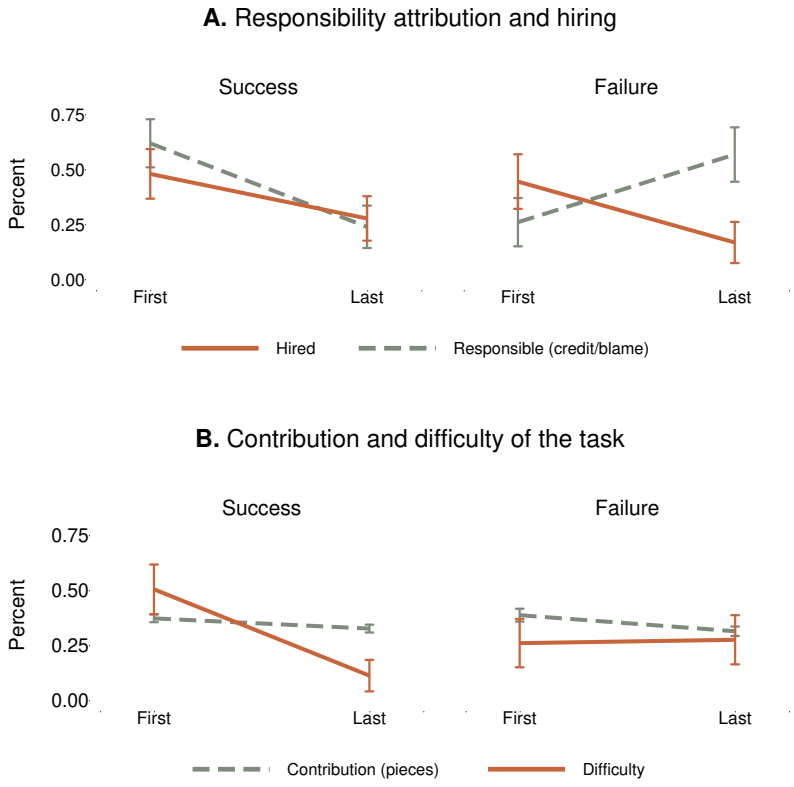


Figure C3 Main Experimental Outcomes: OBJECTIVE. *Notes:* Panel A reports the fraction of managers who hired the first or last builder and the fraction who attributed responsibility for the group outcome to the first or last builder. Panel B reports perceived contribution, measured as the estimated share of pieces placed by each builder, and perceived task difficulty, measured as the fraction of managers who identified the part completed by the corresponding builder as the most difficult. Results are presented separately following success and failure. Vertical bars report 95 percent confidence intervals.

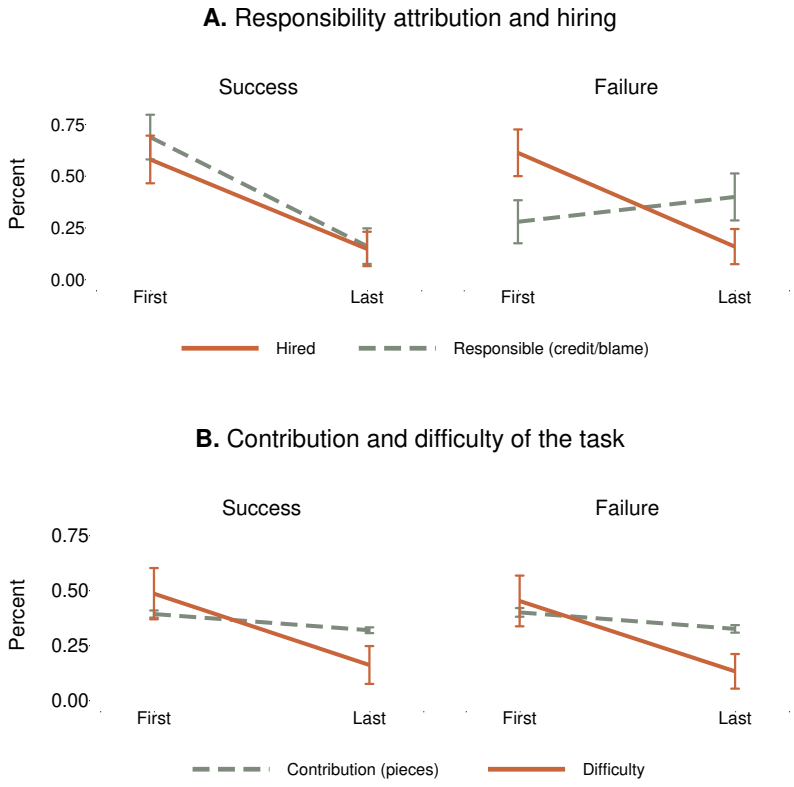


Figure C4 Main Experimental Outcomes: PREOBJECTIVE. *Notes:* Panel A reports the fraction of managers who hired the first or last builder and the fraction who attributed responsibility for the group outcome to the first or last builder. Panel B reports perceived contribution, measured as the estimated share of pieces placed by each builder, and perceived task difficulty, measured as the fraction of managers who identified the part completed by the corresponding builder as the most difficult. Results are presented separately following success and failure. Vertical bars report 95 percent confidence intervals.

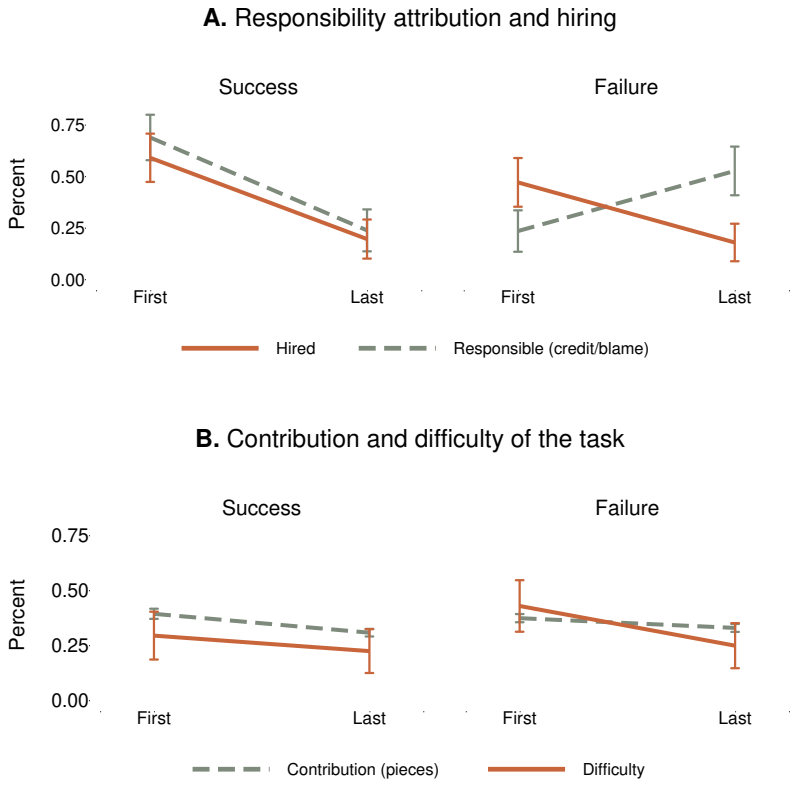


Figure C5 Main Experimental Outcomes: WAGES. *Notes:* Panel A reports the fraction of managers who hired the first or last builder and the fraction who attributed responsibility for the group outcome to the first or last builder. Panel B reports perceived contribution, measured as the estimated share of pieces placed by each builder, and perceived task difficulty, measured as the fraction of managers who identified the part completed by the corresponding builder as the most difficult. Results are presented separately following success and failure. Vertical bars report 95 percent confidence intervals.

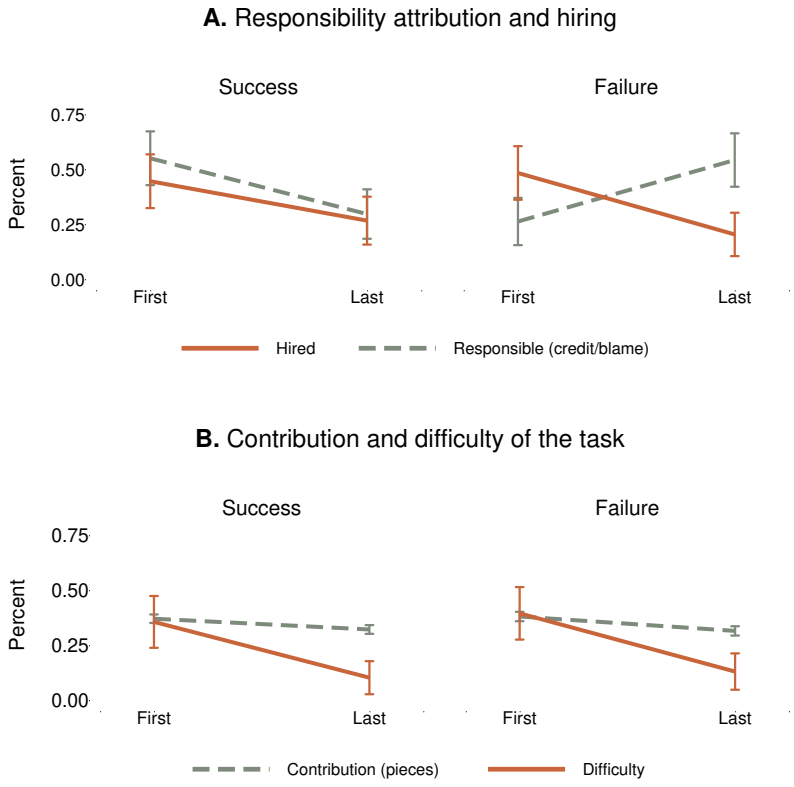


Figure C6 Main Experimental Outcomes: STAKES. *Notes:* Panel A reports the fraction of managers who hired the first or last builder and the fraction who attributed responsibility for the group outcome to the first or last builder. Panel B reports perceived contribution, measured as the estimated share of pieces placed by each builder, and perceived task difficulty, measured as the fraction of managers who identified the part completed by the corresponding builder as the most difficult. Results are presented separately following success and failure. Vertical bars report 95 percent confidence intervals.

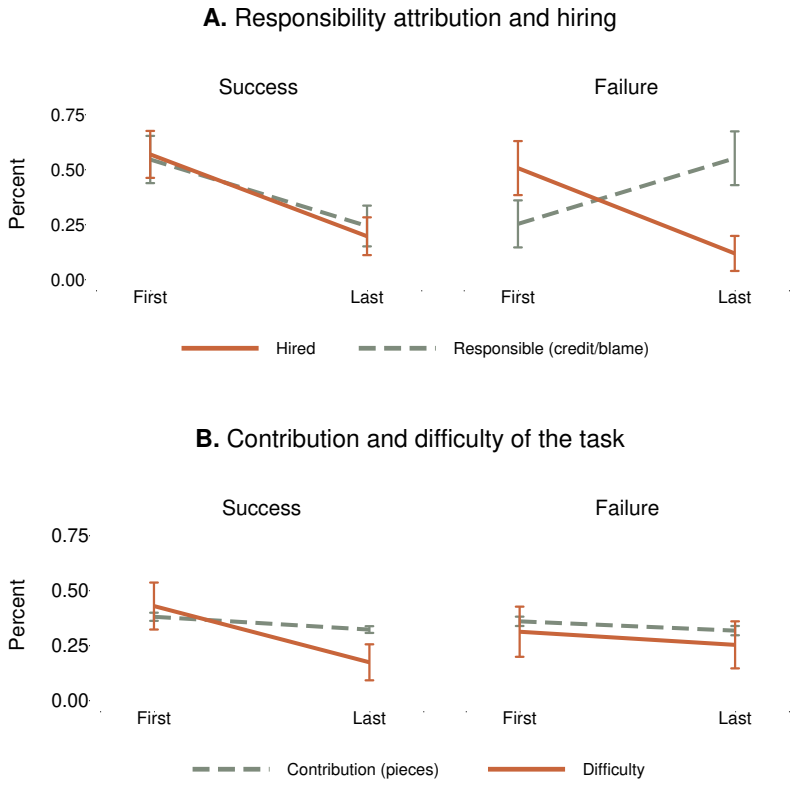


Figure C7 Main Experimental Outcomes: THINK. *Notes:* Panel A reports the fraction of managers who hired the first or last builder and the fraction who attributed responsibility for the group outcome to the first or last builder. Panel B reports perceived contribution, measured as the estimated share of pieces placed by each builder, and perceived task difficulty, measured as the fraction of managers who identified the part completed by the corresponding builder as the most difficult. Results are presented separately following success and failure. Vertical bars report 95 percent confidence intervals.

D Coding Free-Form Responses

D.1 The Coding Task

Following Baranski et al. (2026a), we used the Communication Annotation Tool (CAT) to convert open-ended responses into predefined content categories. Each response was supplied to the large language model (LLM) together with the survey question wording, the relevant study context, and a coding manual specifying category definitions. We used three commercially available LLMs. Each response was classified 5 times with fixed inference settings, and repeated classifications were aggregated using the modal response. CAT automatically checked outputs for missing, invalid, or out-of-range values and retained the coding specification, model settings, call-level outputs, and aggregated classifications, making the procedure transparent and reproducible.

This method has been benchmarked against independent human coding across three published experimental datasets with markedly different free-form text structures to establish reliability. Agreement between individual human–LLM pairs was comparable to agreement between pairs of human coders, and replacing human classifications with LLM classifications preserved the qualitative conclusions of the original statistical analyses. Because our current study is different from the ones analyzed in Baranski et al. (2026a) we test coded 50 observations and found no systematic mistakes in the CAT classifications.

Codebook and Prompt

We provided CAT with the following codebook manual:

1. **Importance**: if the reason is that the subject completed the most important part. Keywords may be “crucial”, “critical” “essential”, “foundation”, etc. It also includes

statements such as “It was the last part”, “Did the finishing touches”, “Started it all” where it is presumed that this was an important part. Enter 1 if the text falls into this category, 0 otherwise.

2. **Difficulty**: if the reason is that the subject completed the most difficult part of the model. Keywords can include “challenging”, “complicated”. Enter 1 if the text falls into this category, 0 otherwise.

3. **Speed**: if the reason is that the subject was the fastest. Other keywords may include “efficient” “quick”, etc. Enter 1 if the text falls into this category, 0 otherwise.

4. **Amount**: If the reason is that the subject placed the highest amount of pieces. Enter 1 if the text falls into this category, 0 otherwise.

5. **Focus**: If the reason is that the subject was very focused, did not get distracted, made no mistakes, was very skillful, had high ability. Enter 1 if the text falls into this category, 0 otherwise.

6. **Random**: when it is stated that the decision was made at random, with no particular motive. Also applies when responders state they “felt”, had a “hunch”, or similar phrasing. Enter 1 if the text falls into this category, 0 otherwise.

7. **Reaction**: the builder reacted by going faster given what she / he was handed. He/she sped up due to the slow progress of those preceding.

8. **Order**: Order=1 only if the response explicitly explains that the builder’s position in the contribution sequence caused a specific consequence and that this consequence is why the respondent selected that builder for hire. The reason must be explicitly associated with being first, middle, or last. Merely mentioning the builder’s position alongside an independent reason–such as speed, skill, pressure, difficulty, focus, or amount contributed–is not sufficient. Do not infer a connection that is not stated.

For a full documentation of the CAT method, see Baranski et al. (2026b).

D.2 Coding and Descriptive Results

Managers provided an open-ended explanation of why they selected a particular builder.[7] In THINK, managers entered this explanation before making the hiring decision and received an additional payment for doing so. In the remaining treatments, managers explained their decision after selecting a builder. We coded each explanation using the eight binary categories described above: *Speed, Focus, Amount completed, Importance of the role, Task difficulty, Random choice or intuition, Reaction and Order*.

The categories are not mutually exclusive. A response may therefore receive more than one code, and the percentages across categories do not sum to 100. The reaction category is omitted from the tables because it appears in only two responses. Table D1 reports the percentage of responses assigned to each of the remaining categories, both in the complete sample and separately according to whether the manager hired the first, middle, or last builder. The final two columns compare managers who hired the first and last builders. The adjusted difference controls for fixed effects for each treatment and outcome combination and for the production video.

Table D1 Managers' Stated Reasons for Hiring

This table reports the percentage of managers whose open-ended hiring explanation was assigned to each coded category. Columns **I–IV** report raw percentages for all managers and separately for managers who hired the first, middle, and last builders. Column **V** reports the adjusted percentage point difference between managers who hired the first and last builders. Column **VI** reports the p-value from a two-sided test that this adjusted difference equals zero. The adjusted comparisons include fixed effects for each treatment and outcome combination and for the production video. Categories are not mutually exclusive, so a response may appear in more than one row. The reaction category is omitted because it was assigned to only two responses.

	I	II	III	IV	V	VI
	ALL MANAGERS	FIRST BUILDER	MIDDLE BUILDER	LAST BUILDER	FIRST − LAST	p-VALUE
Speed	63.6	62.8	67.7	59.2	3.9	0.202
Focus, skill, or no mistakes	43.2	42.0	41.6	48.6	−6.2	0.048
Amount completed	14.9	18.7	10.3	11.9	6.8	0.002
Importance of the role	9.6	13.8	3.5	8.1	5.5	0.004
Task difficulty	11.0	13.4	6.9	11.1	2.7	0.187
Random choice or intuition	5.8	4.9	7.6	5.3	−1.1	0.454
Order	1.9	3.1	0.7	0.6	2.5	< 0.001
Managers	1,811	915	536	360	1,811	

[7] Managers also provided open-ended explanations of their responsibility judgments. However, the analysis in this appendix focuses on explanations of the incentivized hiring decision.

Speed is the most frequently mentioned reason, appearing in 63.6% of all explanations, followed by focus, skill, or the absence of mistakes, which appears in 43.2%. These are the two most common categories among managers who hire the first, middle, or last builder. Managers who hire the first and last builders mention speed at statistically similar rates after accounting for the treatment, outcome, and production video (p = 0.202). Managers who hire the first builder are, however, 6.2 percentage points less likely than managers who hire the last builder to mention focus (p = 0.048). Explicit references to order are uncommon and appear in 1.9% of responses. The other categories are also substantially less frequent than speed and focus, each appearing in no more than 14.9% of explanations.

Some of the less common categories differ across hiring choices. In particular, managers who hire the first builder are 6.8 percentage points more likely than managers who hire the last builder to mention the amount completed and 5.5 percentage points more likely to mention the importance of the builder's role. We report these comparisons descriptively because the coded explanations generally follow the hiring decision and may reflect how managers justify a choice already made.

Table D2 examines whether the adjusted first-minus-last comparisons depend on the treatment in which managers explained their reasoning before deciding or on the inclusion of managers who hired the middle builder. Column **II** excludes THINK. Column **III** restricts the sample to managers who hired either the first or last builder. Column **IV** applies both restrictions.

Table D2 Robustness of Differences in Stated Reasons

This table reports adjusted percentage point differences in coded hiring explanations between managers who hired the first and last builders. Column **I** uses the complete sample. Column **II** excludes THINK, where managers provided their explanation before making the hiring decision. Column **III** restricts the sample to managers who hired either the first or last builder. Column **IV** applies both restrictions. All specifications include fixed effects for each treatment and outcome combination and for the production video. Two-sided robust p-values are reported in parentheses. The reaction category is omitted because it was assigned to only two responses.

	I	**II**	**III**	**IV**
	ALL	NO THINK	FIRST/LAST	FIRST/LAST, NO THINK
Speed	3.9	2.8	3.6	2.3
	(0.202)	(0.390)	(0.253)	(0.476)
Focus, skill, or no mistakes	−6.2	−7.3	−6.3	−7.5
	(0.048)	(0.024)	(0.047)	(0.023)
Amount completed	6.8	6.0	7.6	6.7
	(0.002)	(0.010)	(0.001)	(0.006)
Importance of the role	5.5	5.0	5.6	5.3
	(0.004)	(0.009)	(0.004)	(0.008)
Task difficulty	2.7	4.2	2.8	4.6
	(0.187)	(0.036)	(0.171)	(0.024)
Random choice or intuition	−1.1	−1.5	−1.2	−1.7
	(0.454)	(0.321)	(0.381)	(0.275)
Order	2.5	2.3	2.4	2.2
	(< 0.001)	(< 0.001)	(< 0.001)	(0.001)
Managers	1,811	1,658	1,275	1,167

The broad patterns are unchanged across these specifications. In particular, the adjusted differences in references to speed remain small and statistically insignificant when THINK is excluded and when the sample is restricted to managers who hired either the first or last builder. The differences in references to the amount completed and the importance of the builder's role are also stable across the four specifications, while the mention of focus is significantly lower for managers who hire the first builder. The difference in explicit references to order remains small in magnitude but statistically significant across all four specifications. We therefore emphasize the overall prominence of speed and focus and the rarity of explicit order references, rather than interpreting the smaller differences across categories as evidence about the mechanism producing the first-builder advantage.

E Preregistration, Deviations, and Exploratory Analyses

E.1 Original Preregistered Study

This project began with the study now reported as the pilot in Appendix F. We preregistered that study on AsPredicted before collecting data (AsPredicted #95081). The design, procedures, and results of the original study are reported in Appendix F.

The preregistration established the central question examined throughout the project: whether a contributor's position in a production sequence affects the allocation of credit, blame, and hiring opportunities. It identified the incentivized hiring decision as the primary outcome and responsibility attribution as a secondary outcome. It also specified perceived contribution and perceived task difficulty as additional measures. The original study used four constructed production videos and varied the stated group outcome and the salience of the outcome announcement.

The original study provided evidence of a first-builder advantage in a deliberately stylized environment. Because its production sequences were constructed from independently recorded performances, however, the builders did not participate in an actual sequential production process. The main experiment was designed as a direct extension of this original study. It uses actual three-person production groups and richer production videos while retaining the same central research question, incentivized hiring outcome, and comparison of contributors according to their position in the sequence.

E.2 Relation to the Main Experiment and Additional Treatments

All eight treatments in the main experiment examine the same underlying question: whether the order in which individuals contribute to a common production process affects subsequent hiring decisions. They retain the same three-person choice structure, the same primary behavioral outcome, and the same comparison between earlier and later contributors. The additional treatments change features of the evaluation environment that could attenuate or eliminate the first-builder advantage, including the timing and availability of outcome information, the criterion determining the hiring bonus, the consequences of the hiring decision, and the opportunity for incentivized deliberation.

The additional treatments were not designed around separate directional hypotheses or predictions that the direction of the order effect would reverse. Instead, they examine whether the first-builder advantage persists when particular features of the evaluation environment are changed. We therefore use the preregistered hiring outcome and positional comparison as the common analytical framework across treatments. In this sense, the preregistration applies to the treatment analyses at the level of the research question, primary outcome, and organizing contrast. The specific treatment variations should be understood as extensions of that preregistered framework rather than as separately preregistered interventions.

The BASELINE treatment provides the closest counterpart to the original preregistered study. We first analyze BASELINE separately following success and failure. We then apply the same first-versus-last comparison to each additional treatment. Because the additional treatments were designed to assess the persistence of the same directional pattern rather than to generate distinct predictions about success and failure, we pool the two outcome conditions within each treatment when presenting the treatment-block results. NOINFO, in which managers receive no stated group outcome, enters as a single

condition.

E.3 Deviations from the Preregistered Analysis

The preregistration proposed linear probability models in which hiring and responsibility judgments would be regressed on contributors' positions, with production-video fixed effects and additional specifications controlling for perceived contribution and perceived task difficulty. In the course of organizing the data as a choice problem, we concluded that a conditional-logit model better represents the structure of the hiring decision. Each manager selects exactly one builder from a set of three mutually exclusive alternatives, and the principal explanatory variables and controls vary across builders within that choice set. The conditional-logit specification models these relative choices directly.

This change affects the functional form of the analysis but not the primary outcome or the central comparison between sequential positions. The dependent variable remains the incentivized hiring decision, the first and middle builders are compared with the last builder, and perceived contribution and perceived task difficulty enter as builder-specific controls. We additionally control for observed productivity in the recorded production task.

Production-video indicators and manager characteristics such as age and gender are constant across the three alternatives faced by a given manager and are therefore absorbed by the conditional likelihood. For the same reason, treatment and stated outcome indicators do not enter the conditional-logit models as separate main effects. We estimate the BASELINE models separately following success and failure and estimate separate models for each treatment, allowing the coefficients on position and the builder-specific controls to differ across these conditions.

The preregistration identified responsibility attribution as a secondary outcome. In

the paper, we use responsibility judgments as complementary evidence about the interpretation of the incentivized hiring decision. Specifically, we examine whether hiring and responsibility attribution are positively associated following success and negatively associated following failure. Hiring remains the primary outcome throughout the paper.

E.4 Additional Measures and Exploratory Analyses

The preregistration identified the open-ended explanations of managers' hiring and responsibility judgments as exploratory. It anticipated that the coding categories would be developed after a preliminary reading of the responses and that the analysis might examine whether stated reasons differed according to the contributor selected. Consistent with that designation, Appendix D reports the coded hiring explanations descriptively. We use these responses to characterize the considerations managers report when explaining their choices, not to identify the mechanism producing the first-builder advantage.

The second hiring decision was introduced partway through BASELINE data collection and was not preregistered. We report it only after presenting the primary hiring results and the treatment-block analyses. The analysis is restricted to managers who initially hired the middle builder, for whom the second decision provides a direct choice between the first and last builders. We interpret this analysis as exploratory evidence about whether the first-builder advantage remains visible when the initial decision does not reveal the manager's ranking of those two builders.

The same-video outcome comparison, the analysis separating managers according to the accuracy of their contribution estimates, and the calculation of ex post productivity shortfalls were also not specified in the preregistration. We report these analyses as supporting evidence. The same-video comparison examines the role of the stated group outcome while holding the observed production process fixed. The accuracy analysis examines whether the first-builder advantage is concentrated among managers with

relatively inaccurate perceptions of contribution. The productivity calculations describe the ex post efficiency consequences associated with managers' hiring choices. None of these analyses replaces the preregistered primary hiring outcome or the central comparison between sequential positions.

F Pilot Study: Ex Post Constructed Production Sequences

This appendix reports the original preregistered study described in Appendix E. We refer to it as a pilot because it provided the initial evidence motivating the main experiment. The pilot is particularly informative about the possibility that managers evaluate a late contributor differently because that contributor observed the performance of those who acted earlier and could have adjusted effort in response. In the pilot, the apparent production sequences were constructed from independently recorded performances. The builders therefore did not observe one another and could not respond to the progress of the apparent group.

F.1 Design and Procedure

Builders were undergraduate students recruited from the participant pool of the Social Science Experimental Laboratory at New York University Abu Dhabi. Each builder attended an individual laboratory session and assembled an entire Lego model while being recorded. The model was divided into three consecutive parts containing approximately 40%, 20%, and 40% of the pieces. Each builder completed all three parts and received a fixed participation payment as well as a bonus for completing the full model within the stated time limit.

We constructed apparent three-person production sequences after the individual sessions had been completed. Each sequence combined the first part assembled by one builder, the middle part assembled by a second builder, and the last part assembled by a third builder. Managers were explicitly informed that the videos had been constructed in this way. The videos therefore resembled a sequential production process, although the builders had performed independently and could not observe one another. In partic-

ular, the last builder could not know whether the preceding builders had worked quickly or slowly and had no opportunity to adjust effort in response to their performance.

The videos were edited to retain only periods during which builders were actively placing pieces. Time spent searching for pieces, consulting the instruction manual, or pausing was removed, and the videos were accelerated. These edits limited the performance information available to managers and made the displayed duration of each part uninformative about the builder's actual completion time. The first and last builders completed equal shares of the model, allowing their evaluations to be compared while holding the amount assigned approximately constant.

The pilot used four constructed production videos, two presented as successful and two presented as failed. A production sequence was described as successful when its edited completion time was below three minutes and as failed otherwise. We recruited 390 managers through Prolific. Each manager watched one video and made an incentivized hiring decision. Managers received an additional 2 USD if the builder they hired had completed the full individual building task within the relevant time limit. Managers did not observe the builders' individual completion times.

Managers also identified the builder they considered most responsible for the success or failure of the apparent group, estimated the number of pieces placed by each builder, reported which part appeared most difficult, and provided open-ended explanations for their decisions.

F.2 Results

Table F1 reports conditional logit estimates of the hiring decision using the pooled pilot sample. The last builder is the omitted category. Model 1 includes indicators for the first and middle builders. The coefficient on the first-builder indicator is 0.415 and is statistically significant at the 1 percent level. The coefficient on the middle-builder indicator

is 0.123 and is not statistically significant. The first-builder coefficient is significantly larger than the middle-builder coefficient in this specification ($p = 0.015$).

Model 2 additionally controls for perceived contribution and perceived task difficulty. The first-builder coefficient remains positive and statistically significant and is 0.435. This estimate implies that the conditional odds of hiring the first builder are approximately 1.54 times the odds of hiring the last builder. The middle-builder coefficient is 0.242 and is statistically significant at the 10 percent level. Once the perception controls are included, the coefficients on the first and middle builders are not statistically distinguishable at conventional levels ($p = 0.115$). Perceived contribution is positively and statistically significantly associated with hiring, whereas perceived task difficulty is not statistically significant. Most importantly, accounting for these perceptions does not eliminate the first-builder advantage relative to the last builder.

Table F1 Pooled Hiring Decisions in the Pilot Study

This table reports conditional logit estimates of hiring decisions in the pilot study, pooling successful and failed outcomes. The dependent variable equals one if the manager hires the corresponding builder. The last builder is the omitted category. Column **I** includes indicators for the first and middle builders. Column **II** additionally controls for perceived contribution and perceived task difficulty. The equality-test row reports the p-value from a Wald test of the null hypothesis that the coefficients on the first and middle builders are equal. Standard errors are clustered at the manager level and reported in parentheses. Each manager contributes three builder-level observations. Two managers with incomplete perception measures are omitted from Column **II**. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	I	**II**
	MODEL 1	MODEL 2
First builder	0.415^{***}	0.435^{***}
	(0.125)	(0.128)
Middle builder	0.123	0.242^{*}
	(0.133)	(0.143)
Perceived contribution		2.949^{***}
		(0.765)
Perceived task difficulty		0.109
		(0.134)
Managers	390	388
Observations	1,170	1,164
p-value: First builder = Middle builder	0.015	0.115

F.3 Relation to the Main Experiment

The pilot provided a controlled initial test of whether sequential position affects performance evaluation. Because the apparent production sequences were assembled from independently recorded performances, later builders could not observe the earlier builders and could not react to the apparent group's progress. The pooled first-builder advantage therefore cannot be explained by a belief that the last builder observed poor prior performance and failed to respond by working faster. The result also appears in videos designed to limit observable performance cues, showing that a preference for the first builder can arise even when individual performance is difficult to infer from the production sequence.

The pilot also had important limitations that motivated the experiment reported in the main text. The apparent groups were constructed after builders had completed the task independently rather than generated through an actual sequential production process. The videos deliberately omitted pauses, searches, and consultation of the instructions, thereby removing potentially relevant information about performance. The evidence was also based on only four constructed production videos.

The main experiment addresses these limitations. Builders participate in actual sequential production groups, are randomly assigned to positions, and observe the production process before their own turn. The videos preserve richer information about individual performance, including differences in speed, pauses, and mistakes. The main experiment nevertheless reproduces the first-builder advantage and shows that it persists across changes in outcome information, the hiring criterion, incentives, and opportunities for deliberation.

Taken together, the pilot and the main experiment provide complementary evidence. The pilot shows that the first-builder advantage can arise when later contributors have no opportunity to observe or react to earlier performance and when observable perfor-

mance cues are restricted. The main experiment shows that the advantage persists in an actual sequential production process with richer information and across a wider range of evaluation environments. The combined evidence indicates that sequential position plays a systematic role in performance evaluation.